



US012416028B2

(12) **United States Patent**
Chen et al.

(10) **Patent No.:** **US 12,416,028 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **BIOSYNTHETIC PRODUCTION OF
GAMMA- AND DELTA-LACTONES USING
CYTOCHROME P450 ENZYMES HAVING
SUBTERMINAL HYDROXYLASE ACTIVITY**

FOREIGN PATENT DOCUMENTS

WO	WO 2009/030672	A1	3/2009	
WO	WO-2017137935	A1 *	8/2017	 C12N 15/52
WO	WO 2018/046104	A1	3/2018	
WO	WO 2020/018729	A1	1/2020	

(71) Applicant: **Conagen Inc.**, Bedford, MA (US)

(72) Inventors: **Hui Chen**, North Billerica, MA (US);
Xiaodan Oliver Yu, Lexington, MA (US)

(73) Assignee: **Conagen Inc.**, Bedford, MA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 558 days.

(21) Appl. No.: **17/480,177**

(22) Filed: **Sep. 21, 2021**

(65) **Prior Publication Data**

US 2022/0106618 A1 Apr. 7, 2022

Related U.S. Application Data

(63) Continuation of application No. PCT/US2020/024071, filed on Mar. 21, 2020.

(60) Provisional application No. 62/821,894, filed on Mar. 21, 2019.

(51) **Int. Cl.**
C12P 17/06 (2006.01)
C07D 309/30 (2006.01)
C12N 1/16 (2006.01)
C12N 1/20 (2006.01)
C12N 9/02 (2006.01)
C12P 7/62 (2022.01)
C12P 17/04 (2006.01)

(52) **U.S. Cl.**
CPC **C12P 17/06** (2013.01); **C12N 1/16** (2013.01); **C12N 1/20** (2013.01); **C12P 17/04** (2013.01); **C07D 309/30** (2013.01); **C12N 9/0071** (2013.01); **C12P 7/62** (2013.01)

(58) **Field of Classification Search**
CPC .. C12P 17/06; C12P 17/04; C12P 7/62; C12N 1/16; C12N 1/20; C12N 9/0071; C12N 15/70; C12N 9/0042; C07D 309/30; C12Y 101/01042
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,032,513	A *	7/1991	Page		C12P 17/06
					435/126
7,524,664	B2 *	4/2009	Arnold		C12N 9/0077
					435/71.1
9,970,037	B2	5/2018	Tang et al.		
11,549,131	B2 *	1/2023	Chen		C12P 17/04
2005/0130278	A1	6/2005	Mitsuhashi et al.		
2008/0293101	A1	11/2008	Peters et al.		
2010/0285546	A1	11/2010	Liao et al.		
2021/0261991	A1	8/2021	Chen et al.		

OTHER PUBLICATIONS

Romero-Guido C. et al., "Biochemistry of lactone formation in yeast and fungi and its utilization for the production of flavour and fragrance compounds, Appl Microbiol Biotechnol (2011) 89: pp. 535-547." DOI: 10.1007/s00253-010-2945-0 (Year: 2011).*

Krzyczkowska et al., "Lactone formation in yeast and fungi. Fungal Metabolites, (2017): pp. 461-498." DOI: 10.1007/978-3-319-25001-4_13 (Year: 2017).*

Kitazume et al. "Kinetic analysis of hydroxylation of saturated fatty acids by recombinant P450foxy produced by an *Escherichia coli* expression system. European journal of biochemistry vol. 269,8 (2002): pp 2075-2082." DOI: 10.1046/j.1432-1033.2002.02855.x (Year: 2002).*

Baker et al. "Expression, Purification, and Biochemical Characterization of the Flavocytochrome P450 CYP505A30 from *Myceliophthora thermophila*. ACS omega vol. 2,8 (2017): pp 4705-4724." DOI: 10.1021/acsomega.7b00450 (Year: 2017).*

Nykiel-Szymanska et al., "Elimination and detoxification of 2,4-D by *Umbelopsis isabellina* with the involvement of cytochrome P450, Environ Sci Pollut Res (2018) pp. 2738-2743", submitted with the IDS Dec. 23, 2021 (Year: 2018).*

Berka et al., Comparative genomic analysis of the thermophilic biomass-degrading fungi *Myceliophthora thermophila* and *Thielavia terrestris*. Nat Biotechnol. Oct. 2, 2011;29(10):922-7. doi: 10.1038/nbt.1976.

Dietrich et al., Altering the regioselectivity of the subterminal fatty acid hydroxylase P450 BM-3 towards gamma- and delta-positions. J Biotechnol. Jan. 1, 2009;139(1):115-7. doi: 10.1016/j.jbiotec.2008.10.002. Epub Oct. 15, 2008.

Endrizzi et al., Bioconversion of methyl ricinoleate to 4-hydroxy-decanoic acid and to gamma-decalactone by yeasts of the genus *Candida*. J Basic Microbiol. 1995;35(5):285-92. doi: 10.1002/jobm.3620350503.

Genbank Submission. NCBI; Accession No. ORZ20503, Version ORZ20503.1. cytochrome P450 [*Absidia repens*]. Mondo et al; Apr. 20, 2017. 2 pages.

(Continued)

Primary Examiner — Terry A McKelvey

Assistant Examiner — Andrew T Moehlman

(74) *Attorney, Agent, or Firm* — Wolf, Greenfield & Sacks, P.C.

(57) **ABSTRACT**

Provided herein are methods for making lactones comprising incubating a cellular system expressing a heterologous cytochrome P450 (CYP450) protein with a medium comprising straight chain fatty acids to produce subterminal hydroxy fatty acids, and incubating a yeast cell culture with the subterminal hydroxy fatty acids to produce lactones.

17 Claims, 11 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

OTHER PUBLICATIONS

Genbank Submission. NCBI; Accession No. XP_003663647, version XP_003663647.1. Uncharacterized protein MYCTH_101224 [Thermothelomyces thermophiles ATCC 42464]. Berka et al.; Nov. 27, 2019. 12 Pages.

Nagel et al., Diverging Mechanisms: Cytochrome-P450-Catalyzed Demethylation and γ -Lactone Formation in Bacterial Gibberellin Biosynthesis. *Angew Chem Int Ed Engl.* May 22, 2018;57(21):6082-6085. doi: 10.1002/anie.201713403. Epub Apr. 26, 2018.

Nykiel-Szymańska et al., Elimination and detoxification of 2,4-D by *Umbelopsis isabellina* with the involvement of cytochrome P450. *Environ Sci Pollut Res Int.* Jan. 2018;25(3):2738-2743. doi: 10.1007/s11356-017-0571-4. Epub Nov. 14, 2017.

* cited by examiner

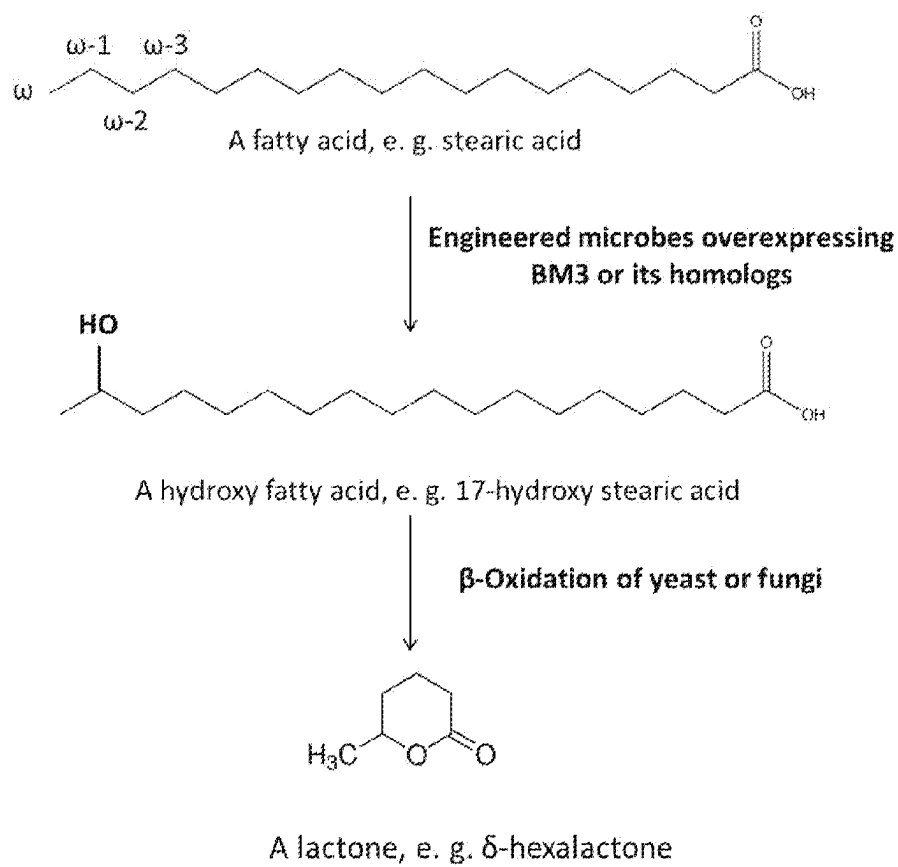


FIG. 1

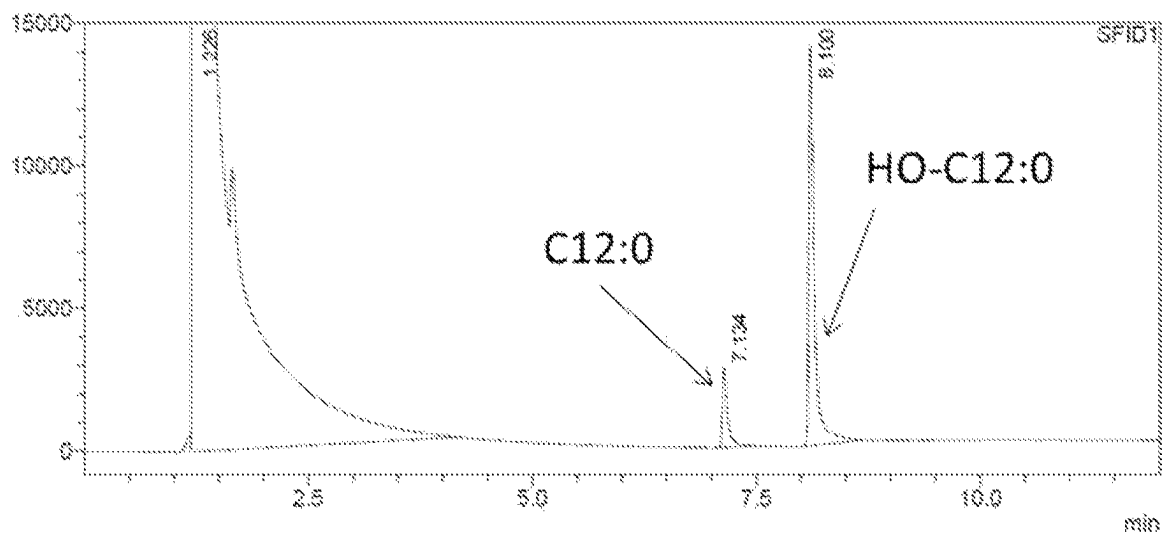


FIG. 2A

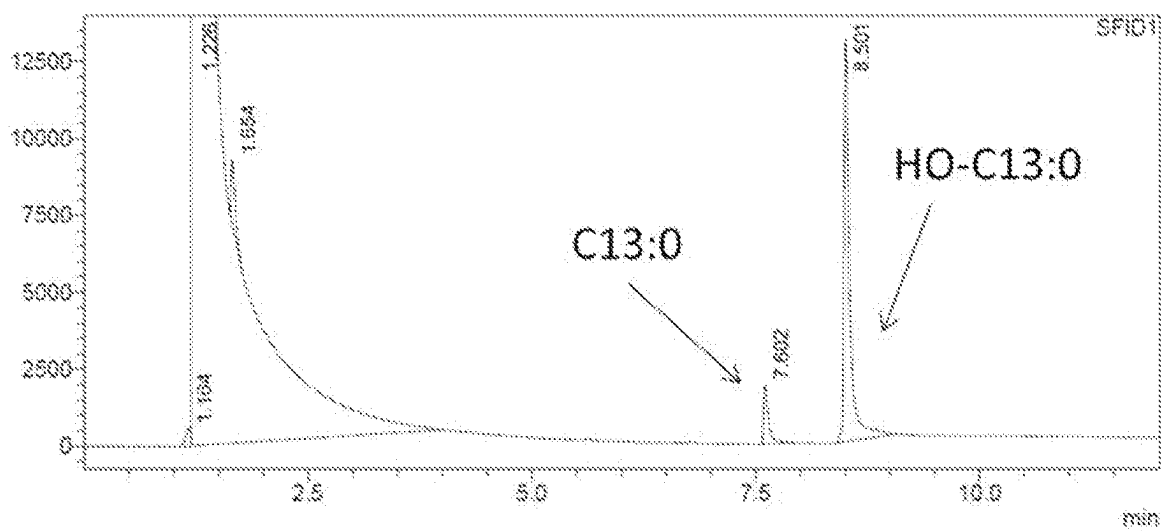


FIG. 2B

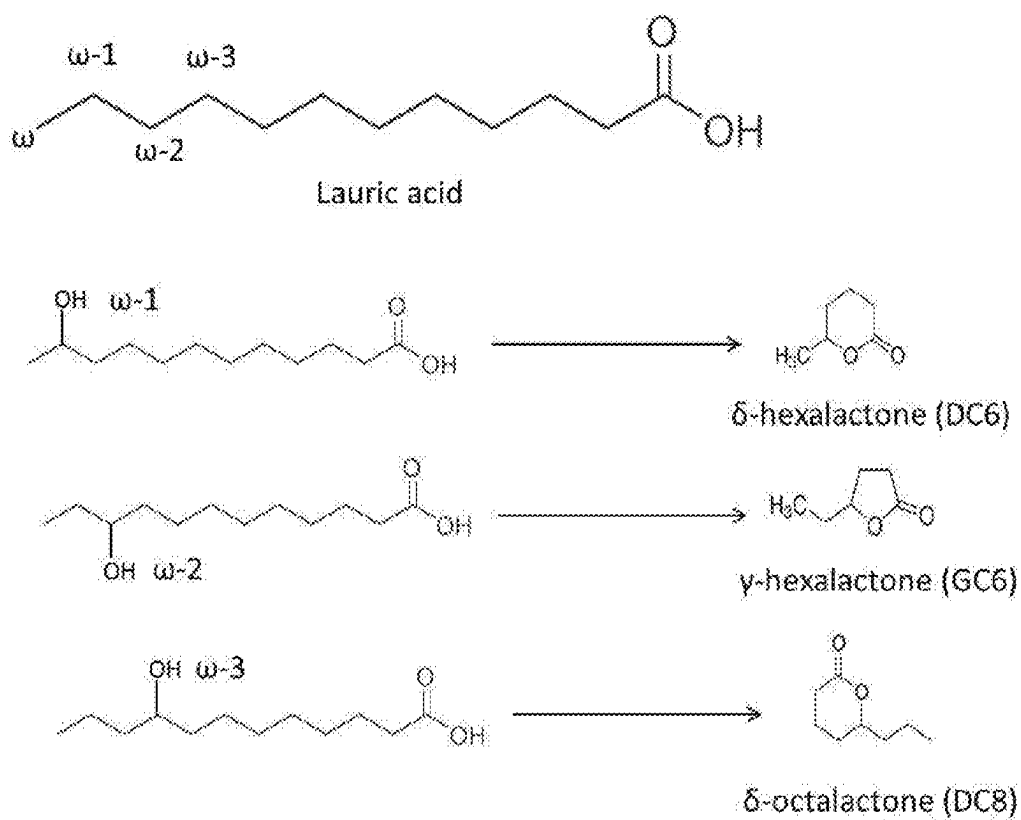


FIG. 3

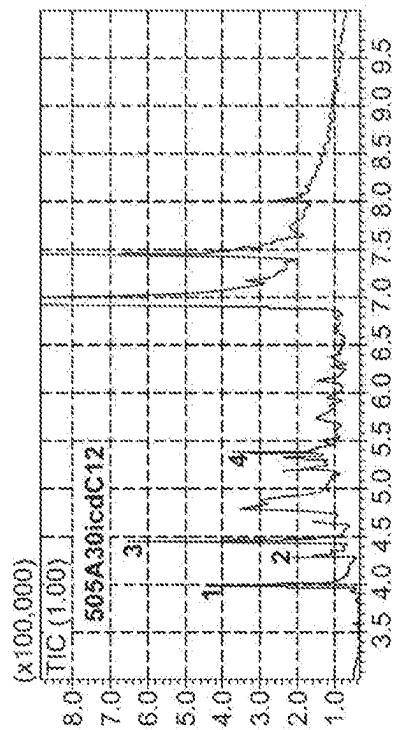


FIG. 4A

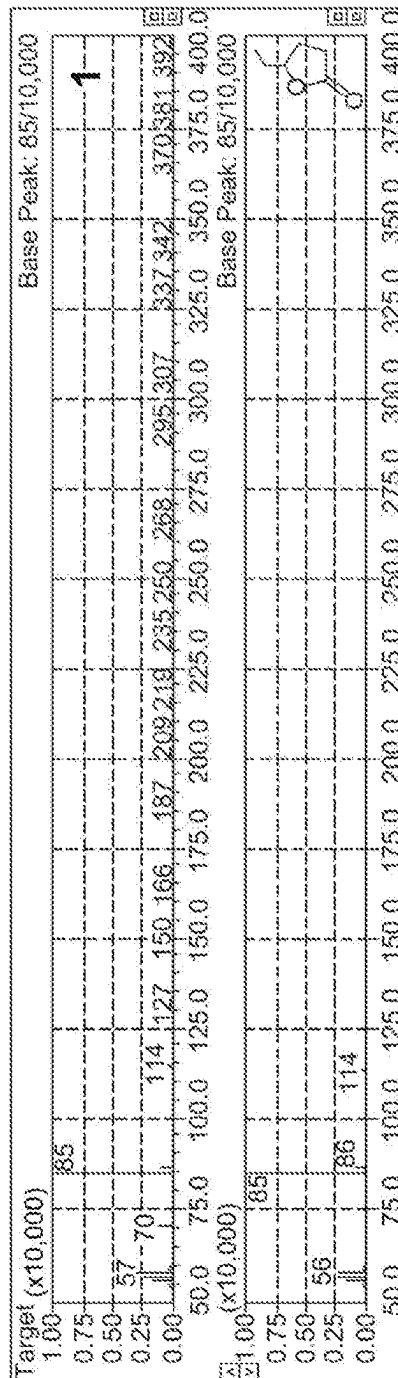


FIG. 4B

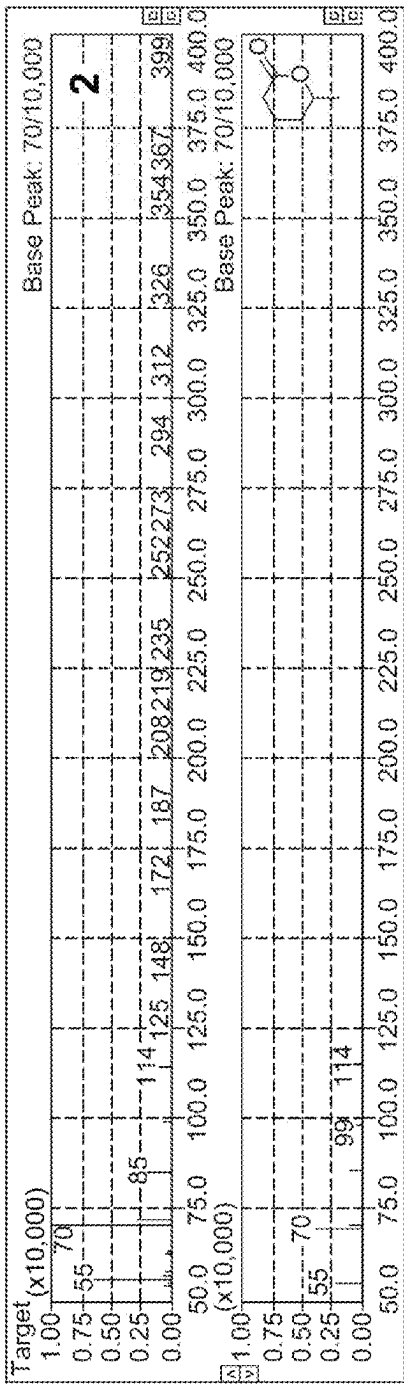


FIG. 4C

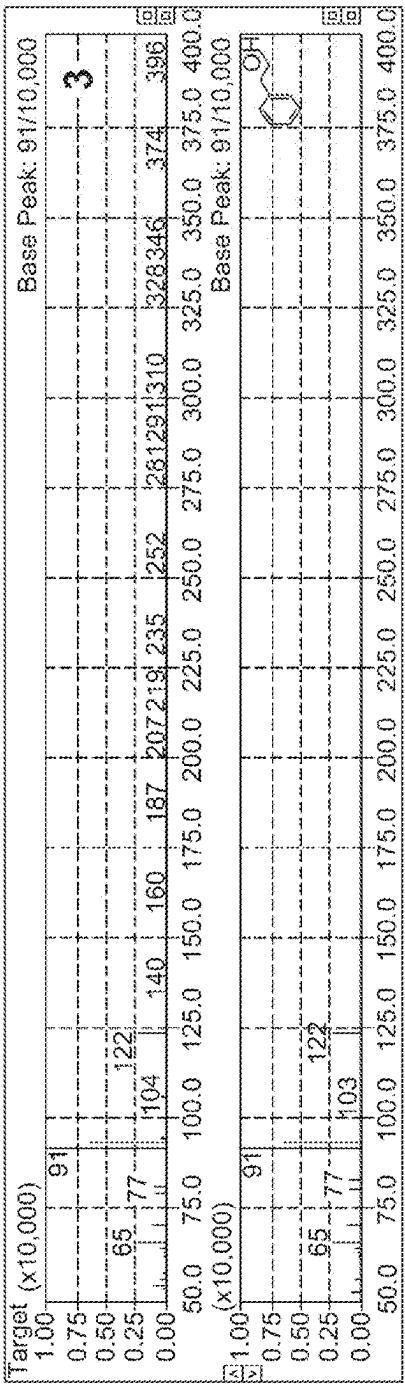


FIG. 4D

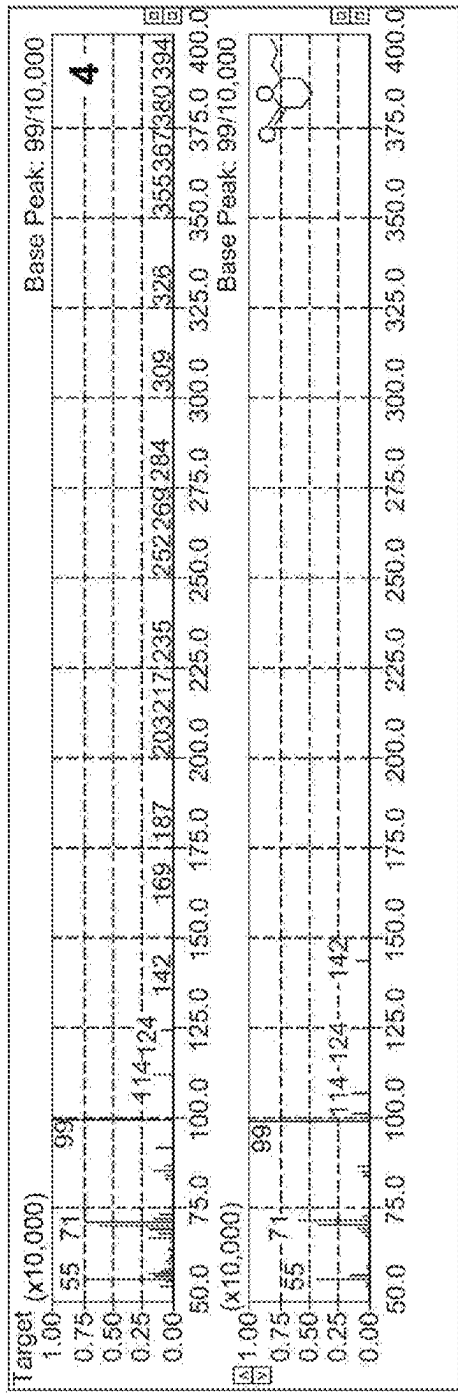


FIG. 4E

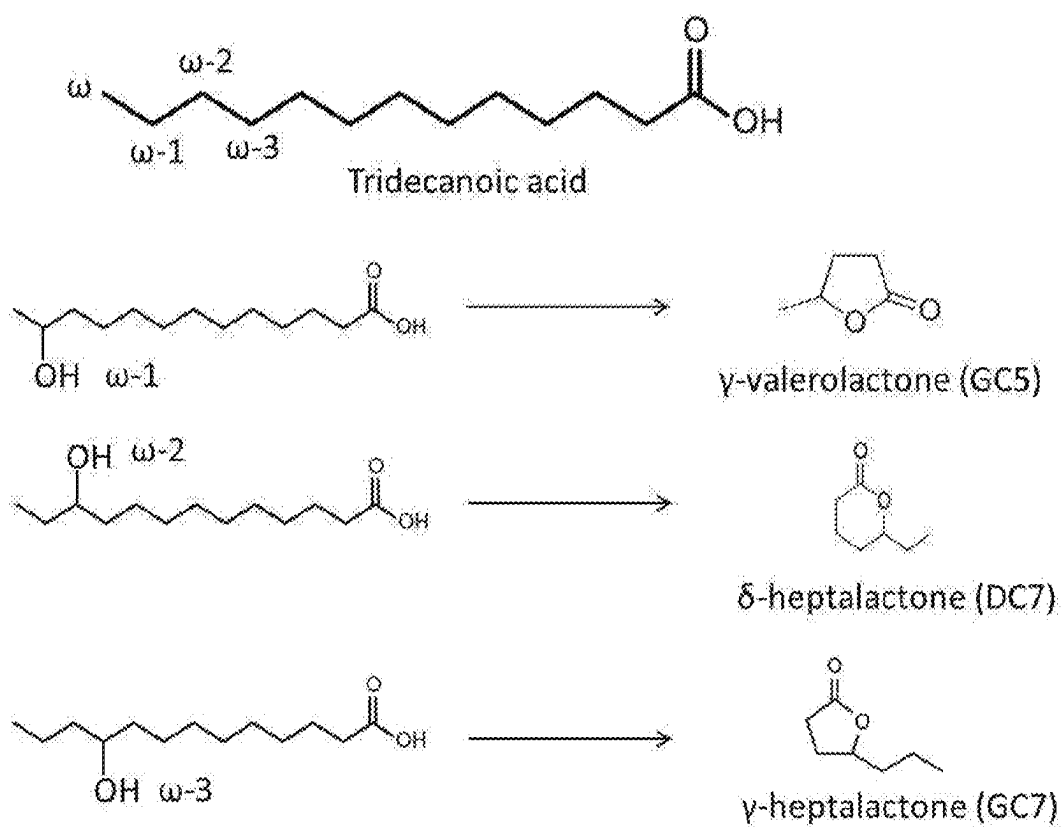


FIG. 5

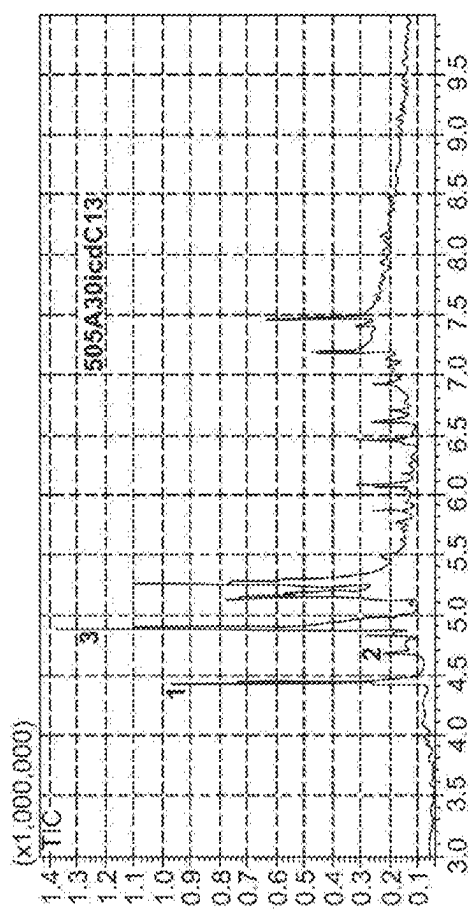


FIG. 6A

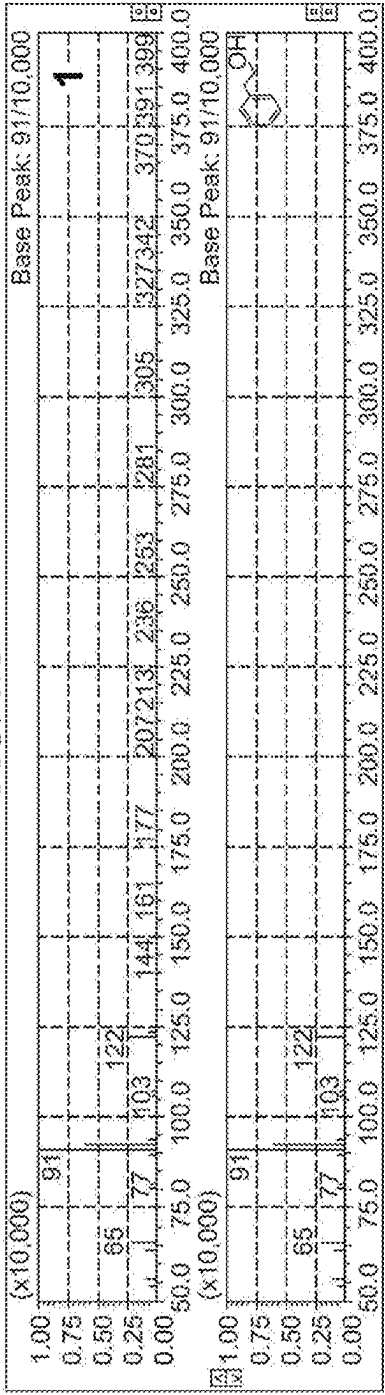


FIG. 6B

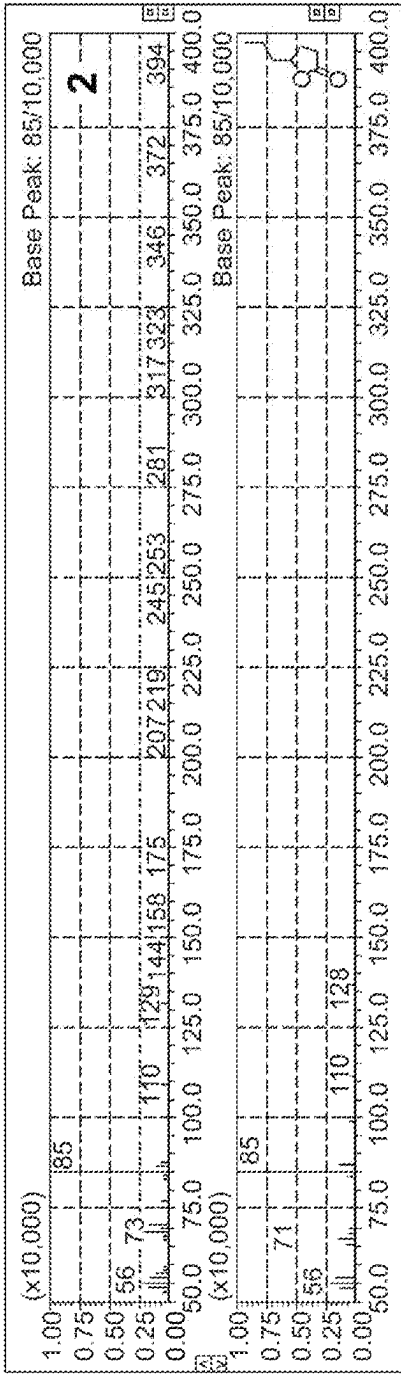


FIG. 6C

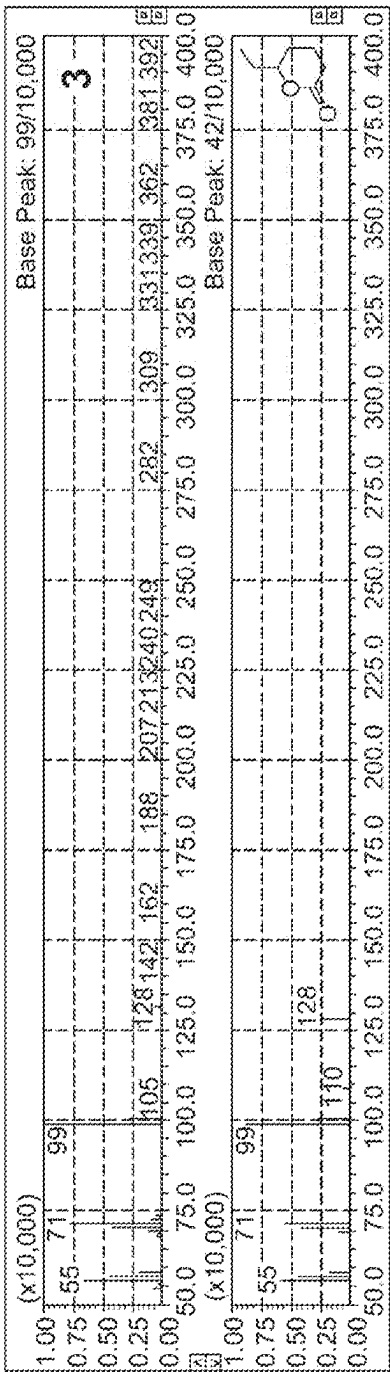


FIG. 6D

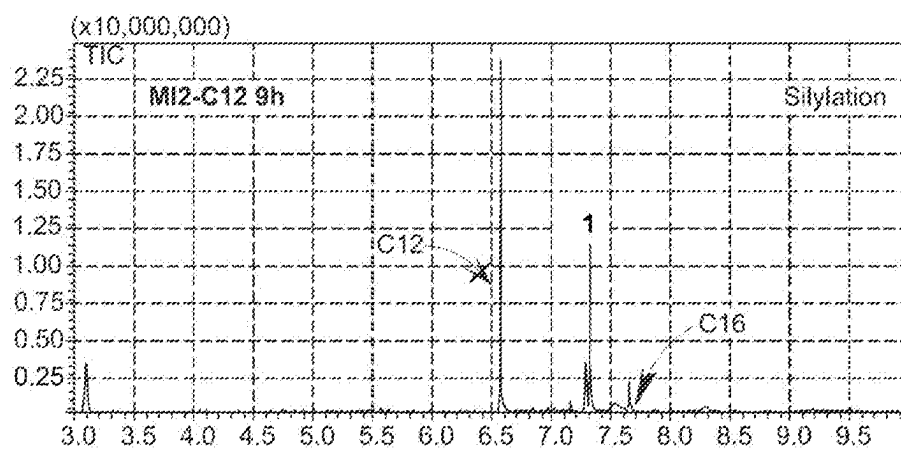


FIG. 7A

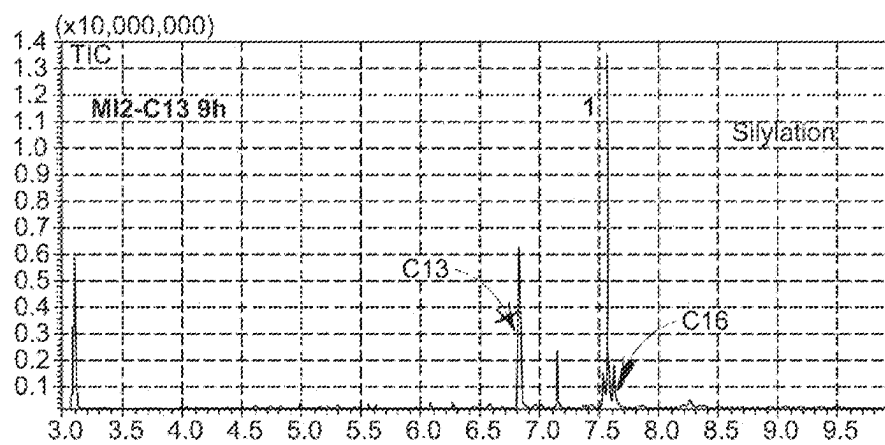


FIG. 7B

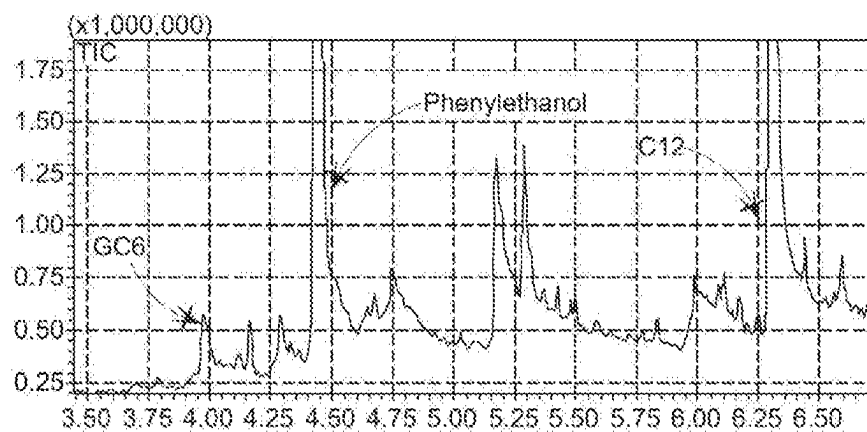


FIG. 8A

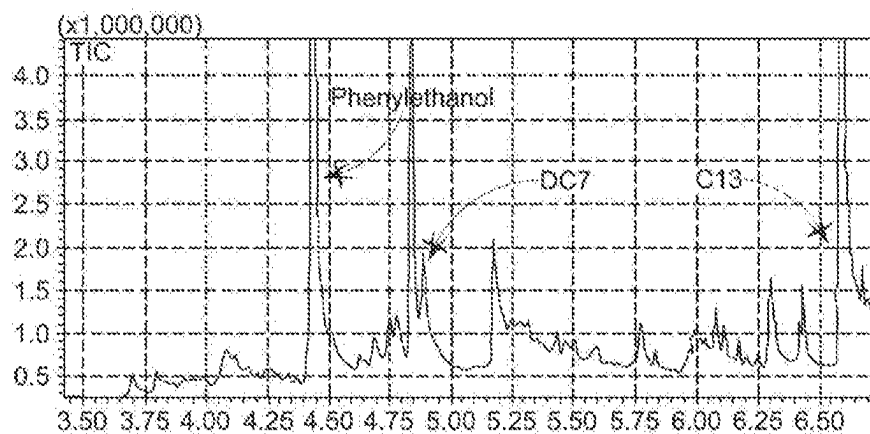


FIG. 8B

1

BIOSYNTHETIC PRODUCTION OF GAMMA- AND DELTA-LACTONES USING CYTOCHROME P450 ENZYMES HAVING SUBTERMINAL HYDROXYLASE ACTIVITY

CROSS-REFERENCE TO RELATED APPLICATIONS

This Application is a Continuation of International Patent Application No. PCT/US2020/024071, filed Mar. 21, 2020, entitled "BIOSYNTHETIC PRODUCTION OF GAMMA- AND DELTA-LACTONES USING CYTOCHROME P450 ENZYMES HAVING SUBTERMINAL HYDROXYLASE ACTIVITY", which claims the priority of the U.S. Provisional Application Ser. No. 62/821,894, filed on Mar. 21, 2019, the entire contents each of which are incorporated herein by reference.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

The contents of the electronic sequence listing (C149770067US01-SEQ-ZJG.xml; Size: 38,878 bytes; and Date of Creation: Sep. 21, 2021) is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

The field of the invention relates to methods and processes for producing gamma- and delta-lactones via modified microbial cultures. More specifically, the present disclosure relates to the production of gamma- and delta-lactones having 5-8 carbon atoms from corresponding fatty acid substrates via enzymatic conversion.

BACKGROUND OF THE INVENTION

There exists a universal desire for foods, fragrances and cosmetics that have pleasant tastes and odors. In many cases, these traits of pleasant smells and tastes are provided through various lactone compounds, including various gamma- and delta-lactones, that possess desirable aroma and flavor characteristics. Current demand for lactone compounds is mainly addressed through either chemical synthesis or extraction processes from plants. Plant extraction-based production has significant disadvantages, such as weather effects on the strength and abundance of the compounds of interest, risk of plant diseases and/or poor harvest, stability of the compound, environmental impact of increased production and trade restrictions. Industrial production can cause environmental damage, may use dangerous precursors and itself may be subject to increased cost due to cost of key substrates.

Accordingly, there is a need in the art for novel methods to produce gamma- and delta-lactones economically and reliably without the limitations posed by plant extraction and chemical synthesis.

SUMMARY OF THE INVENTION

According to the current invention, lactones may be reliably produced at high yield by gene modification and fermentation technology using microorganisms such as bacteria and/or yeasts. These microorganisms are able to synthesize lactones de novo or by biotransformation of fatty acids to provide commercially significant yields. Hence, new production methods are provided herein to reduce costs

2

of gamma- and delta-lactone production and lessen the environmental impact of large-scale cultivation and processing of natural sources from which these lactone compounds can be extracted.

More specifically, the present disclosure encompasses methods and compositions for making lactones by microbial fermentation, wherein the microbial fermentation comprises a cellular system expressing a heterologous cytochrome P450 (CYP450) protein.

The present disclosure is based, in part, on the finding that certain CYP450 proteins and functional variants thereof have hydroxylase activity at specific subterminal carbon atom positions. In particular, certain CYP450 proteins have been identified to cause hydroxylation of straight chain fatty acids at the first (ω -1), second (ω -2), and/or third (ω -3) carbon atoms adjacent to the terminal (ω) carbon atom. By overexpressing these CYP450 proteins in a modified microbial system and feeding straight chain fatty acids as substrates to the said modified microbial system, ω -1, ω -2, and ω -3 hydroxy fatty acids are produced which then can be converted efficiently into various gamma- and delta-lactones at high titers. The present disclosure, therefore, provides economical and reliable methods for producing gamma- and delta-lactones from straight chain fatty acids without the disadvantages associated with chemical synthesis or plant extraction as described above.

Accordingly, one aspect of the present disclosure provides a method for making lactones comprising (a) incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, and incubating the resulting subterminal hydroxy fatty acid with a cell culture followed by acid treatment to produce one or more gamma- and delta-lactones.

In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 90% (e.g., at least 90%, at least 91%, at least 92%, at least 93%, at least 94%) identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila*. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 95% (e.g., at least 95%, at least 96%, at least 97%, at least 98%, at least 99%) identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila*. In some embodiments, the CYP450 protein from *Myceliophthora thermophila* comprises the amino acid sequence of SEQ ID NO: 3.

In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 90% (e.g., at least 90%, at least 91%, at least 92%, at least 93%, at least 94%) identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina*. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 95% (e.g., at least 95%, at least 96%, at least 97%, at least 98%, at least 99%) identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina*. In some embodiments, the CYP450 protein from *Umbelopsis isabellina* comprises the amino acid sequence of SEQ ID NO: 6.

Any methods described herein may further comprise expressing an isocitrate dehydrogenase (Icd) protein (e.g., a heterologous Icd) in the cellular system. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 90% (e.g., at least 90%, at least 91%, at least 92%, at least 93%, at least 94%) identical to the amino acid sequence of Icd from *E. coli*. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 95% (e.g., at least 95%, at least 96%, at least 97%, at least 98%, at least 99%) identical to the amino acid sequence of

Icd from *E. coli*. In some embodiments, the Icd protein from *E. coli* comprises the amino acid sequence of SEQ ID NO: 4.

In some embodiments, the straight chain fatty acid comprises 5 carbon atoms to 40 carbon atoms. In preferred embodiments, the straight chain fatty acid comprises 7 carbon atoms to 20 carbon atoms. In some embodiments, the fatty acid is selected from lauric acid, tridecylic acid, or a combination thereof. In some embodiments, the subterminal hydroxy fatty acid comprises hydroxylated lauric acid, hydroxylated tridecylic acid, or a combination thereof. In some embodiments, the subterminal hydroxy fatty acid comprises a hydroxyl group at a position selected from the group consisting of ω -1, ω -2, and ω -3.

In some embodiments, the cellular system can comprise growing cells. In some embodiments, the cellular system can comprise lyophilized cells. In some embodiments the cellular system can comprise bacterial cells, yeast cells, plant cells that do not naturally produce the lactone of interest, algal cells, fungal cells, or combinations thereof, one or more of which have been transformed to overexpress the protein(s) described herein. In some embodiments, the cellular system can comprise bacteria and/or yeast cells selected from the group consisting of *Escherichia*; *Salmonella*; *Bacillus*; *Acinetobacter*; *Streptomyces*; *Corynebacterium*; *Methylosinus*; *Methylomonas*; *Rhodococcus*; *Pseudomonas*; *Rhodobacter*; *Synechocystis*; *Saccharomyces*; *Zygosaccharomyces*; *Kluyveromyces*; *Candida*; *Hansenula*; *Debaryomyces*; *Mucor*; *Pichia*; *Torulopsis*; *Aspergillus*; *Arthrobotrys*; *Brevibacteria*; *Microbacterium*; *Arthrobacter*; *Citrobacter*; *Klebsiella*; *Mucor*; *Pantoea*; *Corynebacterium*; and *Clostridium*.

In some embodiments, the lactones can comprise five carbons to eight carbons. In some embodiments, the lactones comprise one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone. In some embodiments, the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone.

In some embodiments, the straight chain fatty acid is lauric acid, the subterminal hydroxy fatty acid is hydroxy lauric acid, and the lactones comprises one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone. In some embodiments, the straight chain fatty acid is tridecylic acid, the subterminal hydroxy fatty acid is hydroxy tridecylic acid, and the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone. In some embodiments, the subterminal hydroxy fatty acid is isolated from the cellular system prior to treating it with cell culture.

Any methods described herein may further comprise isolating the lactones from the microbial cell culture or the acidic environment to provide a crude product. In some embodiments, the crude product obtained from such isolating step can include a lactone content that is at least 70% pure. In some embodiments, the method further comprises: i) purifying a crude product comprising the lactones; and, ii) removing solvents under vacuum to provide a concentrated product. In some embodiments, said crude product is purified by column chromatography. In some embodiments, said crude product is purified by acid-base extraction. In some embodiments, said crude product is purified by vacuum distillation. In some embodiments, the method further comprises purifying said lactones using a semi-preparative HPLC.

The disclosure further provides a fermentation product composition comprising a mixture of lactones, wherein the mixture of lactones comprises one or more gamma- and/or delta-lactones having 5 to 8 carbon atoms. The lactone

content in the fermentation product composition can be at least 70% (e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 95%, at least 97%, at least 98%, at least 99% or 100%) of the crude mixture and then can be further purified out of the microbial cell culture.

One skilled in the art will recognize that the lactone composition(s) produced by the methods described herein can be further purified and mixed with other lactones, flavors, or scents to obtain a desired composition for use in a variety of consumer products or foods. For example, a lactone composition described herein can be included in food products (such as beverages, soft drinks, ice cream, dairy products, confectioneries, cereals, chewing gum, baked goods, etc.), dietary supplements, medical nutrition, as well as pharmaceutical products to give desired flavor or aromatic characteristics. In some embodiments, the disclosure provides a consumable product comprising a flavoring or scented amount of a lactone produced by any one of the methods above or described elsewhere herein for use in a specific flavoring context, formula or recipe. In some embodiments, the consumable product can be a food, beverage, perfume, or cosmetic product. In some embodiments, the composition can be selected from beverages, confectioneries, bakery products, cookies, and chewing gums. In certain embodiments, the composition can be a food, beverage, perfume, or cosmetic product having a favor or aroma profile designed to taste or smell like: peach, apricot, pear, maple, coconut, tropical, butterscotch, grenadine and/or date.

While the disclosure is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawing and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

Other features and advantages of this invention will become apparent in the following detailed description of preferred embodiments of this invention, taken with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings form part of the present specification and are included to further demonstrate certain aspects of the present disclosure, which can be better understood by reference to one or more of these drawings in combination with the detailed description of specific embodiments presented herein.

FIG. 1 shows a schematic diagram illustrating an embodiment of the biosynthetic pathway for producing gamma- and delta-lactones according to the present disclosure, in which a C18 fatty acid (stearic acid) is bioconverted into delta-hexalactone via a 2-step process involving the subterminal hydroxy fatty acid (17-hydroxy stearic acid) as an intermediate.

FIG. 2A shows GC/MS spectra confirming production of subterminal hydroxy fatty acid HO-C12:0 from the corresponding straight chain fatty acid C12:0 in a cellular system in which *M. thermophile* CYP505A30 and *E. coli* icd were overexpressed. Samples were taken 4 hours after bioconversion.

5

FIG. 2B shows GC/MS spectra confirming production of subterminal hydroxy fatty acid HO-C13:0 from the corresponding straight chain fatty acid C13:0 in a cellular system in which *M. thermophile* CYP505A30 and *E. coli* icd were overexpressed. Samples were taken 4 hours after bioconversion.

FIG. 3 illustrates the production of δ -hexalactone (DC6), γ -hexalactone (GC6), and 8-octalactone (DC8) from lauric acid (C12:0).

FIGS. 4A, 4B, 4C, 4D, and 4E show GC/MS spectra confirming production of gamma- and delta-lactones from subterminal hydroxy fatty acids HO-12:0. FIG. 4A is at the top. FIG. 4B is the second from top. FIG. 4C is in the middle. FIG. 4D is the fourth from top and FIG. 4E is at the bottom. Peaks are identified based on the retention times and mass spectra of the peaks compared with standards from Sigma. FIG. 4B and peak 1: γ -hexalactone (GC6); FIG. 4C and peak 2: δ -hexalactone (DC6); FIG. 4D and peak 3: phenethyl alcohol; and FIG. 4E and peak 4: δ -octalactone (DC8).

FIG. 5 illustrates the production of γ -valerolactone (GC5), δ -heptalactone (DC7), and γ -heptalactone (GC7) from tridecanoic acid (C13:0).

FIGS. 6A, 6B, 6C, and 6D show GC/MS spectra confirming production of gamma- and delta-lactones from subterminal hydroxy fatty acids HO-13:0. Peaks are identified based on the retention times and mass spectra of the peaks compared with standards from Sigma. FIG. 6A is at the top. FIG. 6B is the second from top. FIG. 6C is the third from the top and FIG. 6D is at the bottom. FIG. 6B and peak 1: phenethyl alcohol; FIG. 6C and peak 2: γ -heptalactone (GC7); FIG. 6D and peak 3: δ -heptalactone (DC7).

FIGS. 7A and 7B show GC/MS spectra confirming production of subterminal hydroxy fatty acid HO-C12:0 (peak 1 in FIG. 7A) and HO-C13:0 (peak 1 in FIG. 7B) from C12:0 and C13:0, respectively, in a cellular system in which the gene MI2 was overexpressed. Samples were taken 9 h after the bioconversion and silylated with N-Triethylsilyl-N-methyl trifluoroacetamide.

FIGS. 8A and 8B show GC/MS spectra confirming lactone production from different hydroxy fatty acids. FIG. 8A: *Candida boidinii* culture grown in MI2-C12-CSL medium. FIG. 8B: *Candida boidinii* culture grown in MI2-C13-CSL medium. Abbreviations in FIGS. 8A-8B have the following meaning: GC6: γ -hexalactone; DC7: δ -heptalactone (DC7); C12: lauric acid; C13: tridecanoic acid. Samples were taken 2 days after inoculation.

DESCRIPTION OF THE PREFERRED EMBODIMENT

Lactones are industrially important flavor compounds that are widely found in foods, fruits, and beverages and are used in many fruity aromatic foods and cosmetics (see, e.g., Romero-Guido et al., 2011. Biochemistry of lactone formation in yeast and fungi and its utilization for the production of flavor and fragrance compounds. Appl. Microbiol. Biotechnol. 89:535-547). The present disclosure provides, in some embodiments, methods of making lactones from fatty acids. In some embodiments, the method comprises a first step, in which a cellular system expressing a heterologous cytochrome P450 (CYP450) protein converts fatty acids to hydroxy fatty acids. In some embodiments, the method comprises a second step, in which a microbial cell culture, e.g., a yeast cell culture, shorten the length of subterminal hydroxy fatty acids through a biochemical process called β -oxidation to appropriate length which upon cyclization

6

yield appropriate lactones. Thus, the length of the subterminal hydroxy acids at the end of the second step of the process according to the present invention is shorter than the length of the subterminal hydroxy fatty acids obtained at the end of the first step of the process according to the present invention. In some embodiments, the second step further comprises an acidification step which induces the cyclization of the shortened subterminal hydroxy fatty acids to appropriate lactones. Methods provided herein further comprise isolating the lactones from the cell culture or the acidic environment.

Methods of Making Lactones

Methods described herein, in some embodiments, provide for the production of lactones from fatty acids. In some embodiments, methods of making lactones comprise (a) a step of incubating a cellular system expressing a heterologous cytochrome P450 (CYP450) protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, (b) a step of incubating the subterminal hydroxyl fatty acid from step (a) with a microbial cell culture to shorten the length of the subterminal hydroxy acid to appropriate length and (c) a step of subjecting the shortened subterminal hydroxy acid from step (b) to cyclization in an acidic environment to obtain appropriate lactone.

In some embodiments, methods of making lactones comprise incubating cell pellets harvested from the cellular system expressing the heterologous CYP450 protein with a medium comprising fatty acids. In some embodiments, methods of making lactones comprise incubating resuspended cell pellets harvested from the cellular system with the medium comprising fatty acids.

Methods of making lactones described herein comprise incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids at any ratio. In some embodiments, methods of making lactones comprise incubating a cellular system comprising between 1 gram of cells per liter to 200 grams of cells per liter (1 g/L to 200 g/L) and a medium comprising between 1 gram of fatty acids per liter to 20 grams of fatty acids per liter (1 g/L to 20 g/L). In some embodiments, methods of making lactones comprise incubating a cellular system comprising 100 grams of cells per liter (100 g/L) and a medium comprising 3 grams of fatty acids per liter (3 g/L).

In some embodiments, methods of making lactones comprise incubating a cellular system with a medium comprising fatty acids for a suitable period sufficient for formation of hydroxy fatty acids. In some embodiments, methods of making lactones comprise contacting the hydroxy fatty acids with a cell culture for a suitable period sufficient for formation of lactones in the microbial cell culture. In some embodiments, methods of making lactones comprise isolating the hydroxy fatty acids from the cellular system prior to contact with the microbial cell culture. In some embodiments, methods of making lactones comprise incubating the hydroxy fatty acids in a medium with the microbial cell culture.

Methods of making lactones described herein encompass incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids for a specific period of time. In some embodiments, methods of making lactones comprise incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids for a period of time between 0.5 hour to 96 hours.

Methods of making lactones described herein encompass incubating a cellular system expressing a heterologous

CYP450 protein with a medium comprising fatty acids at a specified temperature. In some embodiments, methods of making lactones comprise incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids at a temperature of 16° C. to 40° C. Cytochrome P450

Cytochrome P450 (CYP450) proteins, generally, are enzymes that catalyze oxidation of substrates. CYP450 proteins are members of the hemeprotein superfamily, and contain heme as a cofactor. CYP450 enzymes have been identified in many organisms including, but not limited to, animals, plants, fungi, bacteria, archaea, and viruses. Any CYP450 protein that catalyzes subterminal hydroxylation of fatty acids may be used in methods described herein, or any functional variant or fragment thereof. Examples of CYP450 proteins that have been reported to have subterminal hydroxylase activity include, but are not limited to, CYP102A1 (or P450_{BMS}) from *Bacillus megaterium*, CYP450 protein from *Myceliophthora thermophila* (Uniprot #G2QDZ3), CYP450 from *Fusarium oxysporum*, CYP450 protein from *Umbelopsis isabellina*, CYP147G1 from *Mycobacterium marinum*, CYP109D1 from *Sorangium cellulosum*, CYP709C1 from *Triticum aestivum*, CYP109B1 from *Bacillus subtilis*, CYP505D6 from *Phanerochaete chrysosporium*, or any functional variant or fragment thereof. In some embodiments, more than one CYP450 protein is used in a method described herein. In some embodiments, 2, 3, 4, or 5 different CYP450 proteins are used in a method described herein.

In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila* encoded by the nucleic acid sequence of SEQ ID NO: 1. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Myceliophthora thermophila* encoded by the nucleic acid sequence of SEQ ID NO: 1. In some embodiments, the CYP450 protein comprises an amino acid sequence encoded by nucleic acids in SEQ ID NO: 1.

In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila* provided in SEQ ID NO: 3. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Myceliophthora thermophila* provided in SEQ ID NO: 3. In some embodiments, the CYP450 protein comprises the amino acid sequence of SEQ ID NO: 3.

In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina* encoded by the nucleic acid sequence of SEQ ID NO: 5. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Umbelopsis isabellina* encoded by the nucleic acid sequence of SEQ ID NO: 5. In some embodiments, the CYP450 protein comprises an amino acid sequence encoded by nucleic acids in SEQ ID NO: 5.

In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the

amino acid sequence of CYP450 from *Umbelopsis isabellina* provided in SEQ ID NO: 6. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Umbelopsis isabellina* provided in SEQ ID NO: 6. In some embodiments, the CYP450 protein comprises the amino acid sequence of SEQ ID NO: 6.

CYP450 proteins, in some embodiments, catalyze conversion of fatty acids to hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -1 to ω -3 hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -1 hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -2 hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -3 hydroxy fatty acids.

Isocitrate Dehydrogenase (Icd)

Isocitrate dehydrogenase (Icd) proteins, generally, are enzymes that catalyze oxidative decarboxylation of isocitrate. Icd proteins have been identified in many organisms including, but not limited to, animals, plants, fungi, bacteria, archaea, and viruses. Icd proteins, in some embodiments, convert NAD⁺ to NADH or NADP⁺ to NADPH. In some embodiments, NADH and/or NADPH generated by an Icd protein may be utilized by another enzyme (e.g., CYP450), thus promoting the activity of the other enzyme. Any Icd protein that catalyzes conversion of NAD⁺ to NADH or NADP⁺ to NADPH may be used in methods described herein, or any functional variant or fragment thereof. An example of an Icd protein includes, but is not limited to, Icd protein from *E. coli* (Uniprot #P08200). In some embodiments, more than one Icd protein is used in a method described herein. In some embodiments, 2, 3, 4, or 5 different Icd proteins are used in a method described herein.

In some embodiments, the Icd protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of Icd from *E. coli* encoded by the nucleic acid sequence of SEQ ID NO: 2. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of Icd protein from *E. coli* encoded by the nucleic acid sequence of SEQ ID NO: 2. In some embodiments, the Icd protein comprises an amino acid sequence encoded by nucleic acids in SEQ ID NO: 2.

In some embodiments, the Icd protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of Icd from *E. coli* provided in SEQ ID NO: 4. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of Icd protein from *E. coli* provided in SEQ ID NO: 4. In some embodiments, the Icd protein comprises the amino acid sequence of SEQ ID NO: 4.

Lactones

According to methods described herein, hydroxy fatty acids may be converted to lactones. According to the present disclosure, the methods provided herein convert subterminal hydroxy fatty acids into γ -lactones, δ -lactones, and combinations thereof. Examples of lactones include, but are not limited to, δ -hexalactone, γ -hexalactone, δ -octalactone, γ -valerolactone, δ -heptalactone, and γ -heptalactone.

In some embodiments, subterminal hydroxy lauric acid is converted to δ -hexalactone, γ -hexalactone, and/or δ -octalac-

tone. In some embodiments, hydroxy tridecanoic acid is converted to γ -valerolactone, δ -heptalactone, and/or γ -heptalactone.

Lactones described herein, in some embodiments, comprise 5 carbons to 8 carbons. In some embodiments, the lactones comprise 5 carbons. In some embodiments, the lactones comprise 6 carbons. In some embodiments, the lactones comprise 7 carbons. In some embodiments, the lactones comprise 8 carbons.

Any lactones produced according to methods described herein may be isolated or purified from a cell culture using any method known in the art. In some embodiments, lactones are isolated or purified from the cell culture with chromatography. In some embodiments, lactones are isolated or purified from the cell culture with organic solvent. In some embodiments, the isolated or purified lactones can be incorporated into a product, e.g., a food, a beverage, a perfume or a cosmetic.

Fatty Acids

According to methods described herein, fatty acids may be converted to hydroxy fatty acids. Any fatty acid may be converted to its hydroxylated form in accordance with methods disclosed herein. Examples of fatty acids include, but are not limited to, pentanoic acid, hexanoic acid, heptanoic acid, octanoic acid, nonanoic acid, decanoic acid, undecanoic acid, dodecanoic acid, tridecanoic acid, tetradecanoic acid, pentadecanoic acid, hexadecanoic acid, heptadecanoic acid, octadecanoic acid, nonadecanoic acid, eicosanoic acid, heneicosanoic acid, docosanoic acid, tricosanoic acid, tetracosanoic acid, pentacosanoic acid, hexacosanoic acid, heptacosanoic acid, octacosanoic acid, nonacosanoic acid, triacontanoic acid, henatriacontanoic acid, dotriacontanoic acid, tritriacontanoic acid, tetratriacontanoic acid, pentatriacontanoic acid, hexatriacontanoic acid, heptatriacontanoic acid, octatriacontanoic acid, nonatriacontanoic acid, and tetracontanoic acid. Table 1 provides common names, structural formulas, and lipid numbers of exemplary fatty acids.

Fatty acids described herein, in some embodiments, comprise 5 carbons to 40 carbons. In some embodiments, the fatty acids comprise 5 carbons, 6 carbons, 7 carbons, 8 carbons, 9 carbons, 10 carbons, 11 carbons, 12 carbons, 13 carbons, 14 carbons, 15 carbons, 16 carbons, 17 carbons, 18 carbons, 19 carbons, 20 carbons, 21 carbons, 22 carbons, 23 carbons, 24 carbons, 25 carbons, 26 carbons, 27 carbons, 28 carbons, 29 carbons, 30 carbons, 31 carbons, 32 carbons, 33 carbons, 34 carbons, 35 carbons, 36 carbons, 37 carbons, 38 carbons, 39 carbons, 40 carbons, or more carbons.

Fatty acids described herein, in some embodiments, are provided in a medium. In some embodiments, the medium comprising fatty acids comprises a cell culture medium. In some embodiments, the medium comprising fatty acids comprises salts. Salts are combinations of anions and cations. Non-limiting examples of cations include lithium, sodium, potassium magnesium, calcium, and ammonium. Non-limiting examples of anions include acetate, chloride, sulfate, and phosphate. In some embodiments, the medium comprising fatty acids comprises isocitric acid trisodium salt.

In some embodiments, the medium comprising fatty acids comprises a buffer. Examples of buffers include, without limitation, phosphate buffer, Tris buffer, MOPS buffer, HEPES buffer, citrate buffer, acetate buffer, malate buffer, MES buffer, histidine buffer, PIPES buffer, bis-tris buffer, and ethanolamine buffer.

In some embodiments, the medium comprising fatty acids comprises a detergent. Non-limiting examples of detergents

include polysorbate 20 (TWEEN® 20), polyoxyethylene-sorbitan monopalmitate (TWEEN® 40), 4-(1,1,3,3-Tetramethylbutyl)phenyl-polyethylene glycol (TRITON™ X-100), sodium dodecyl sulfate (SDS), ethyl trimethylammonium bromide (ETMAB), lauryl trimethyl ammonium bromide (LTAB), and lauryl trimethylammonium chloride (LTAC).

A medium comprising fatty acids may comprise any concentration of fatty acids. In some embodiments, the medium comprising fatty acids comprises 0.5 g/L to 100 g/L fatty acids, e.g., 1 g/L to 10 g/L fatty acids. In some embodiments, the medium comprising fatty acids comprises 0.5 g/L, 1 g/L, 2 g/L, 3 g/L, 4 g/L, 5 g/L, 6 g/L, 7 g/L, 8 g/L, 9 g/L, 10 g/L, 20 g/L, 30 g/L, 40 g/L, 50 g/L, 60 g/L, 70 g/L, 80 g/L, 90 g/L or 100 g/L fatty acids.

Any fatty acid described herein may be hydroxylated according to methods described herein. In some embodiments, pentanoic acid is converted to hydroxylated pentanoic acid, hexanoic acid is converted to hydroxylated hexanoic acid, heptanoic acid is converted to hydroxylated heptanoic acid, octanoic acid is converted to hydroxylated octanoic acid, nonanoic acid is converted to hydroxylated nonanoic acid, decanoic acid is converted to hydroxylated decanoic acid, undecanoic acid is converted to hydroxylated undecanoic acid, dodecanoic acid is converted to hydroxylated dodecanoic acid, tridecanoic acid is converted to hydroxylated tridecanoic acid, tetradecanoic acid is converted to hydroxylated tetradecanoic acid, pentadecanoic acid is converted to hydroxylated pentadecanoic acid, hexadecanoic acid is converted to hydroxylated hexadecanoic acid, heptadecanoic acid is converted to hydroxylated heptadecanoic acid, octadecanoic acid is converted to hydroxylated octadecanoic acid, nonadecanoic acid is converted to hydroxylated nonadecanoic acid, eicosanoic acid is converted to hydroxylated eicosanoic acid, heneicosanoic acid is converted to hydroxylated heneicosanoic acid, docosanoic acid is converted to hydroxylated docosanoic acid, tricosanoic acid is converted to hydroxylated tricosanoic acid, tetracosanoic acid is converted to hydroxylated tetracosanoic acid, pentacosanoic acid is converted to hydroxylated pentacosanoic acid, hexacosanoic acid is converted to hydroxylated hexacosanoic acid, heptacosanoic acid is converted to hydroxylated heptacosanoic acid, octacosanoic acid is converted to hydroxylated octacosanoic acid, nonacosanoic acid is converted to hydroxylated nonacosanoic acid, triacontanoic acid is converted to hydroxylated triacontanoic acid, henatriacontanoic acid is converted to hydroxylated henatriacontanoic acid, dotriacontanoic acid is converted to hydroxylated dotriacontanoic acid, tritriacontanoic acid is converted to hydroxylated tritriacontanoic acid, tetratriacontanoic acid is converted to hydroxylated tetratriacontanoic acid, pentatriacontanoic acid is converted to hydroxylated pentatriacontanoic acid, hexatriacontanoic acid is converted to hydroxylated hexatriacontanoic acid, heptatriacontanoic acid is converted to hydroxylated heptatriacontanoic acid, octatriacontanoic acid is converted to hydroxylated octatriacontanoic acid, nonatriacontanoic acid is converted to hydroxylated nonatriacontanoic acid, and tetracontanoic acid is converted to hydroxylated tetracontanoic acid.

Hydroxy fatty acids, in some embodiments, comprise a hydroxyl group at position ω -1, ω -2, ω -3, or combinations thereof. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω -1, ω -2, and ω -3. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω -1. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position

ω-2. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω-3.

In some embodiments, hydroxy fatty acids are contacted with a cell culture. In some embodiments, contacting the hydroxy fatty acids with a cell culture comprises exposing a hydroxy fatty acid with a cell culture for a suitable period sufficient for formation of lactones in the cell culture, if any. In some embodiments, the hydroxy fatty acids are isolated prior to contact with the cell culture. In some embodiments, the hydroxy fatty acids in a medium are contacted with the cell culture.

Cellular Systems

A cellular system refers to any cell or cells that provide for expression of ectopic proteins or heterologous proteins (e.g., CYP450). A cellular system includes, but is not limited to, bacterial cells, yeast cells, plant cells, and animal cells. In some embodiments, the cellular system comprises bacterial cells, yeast cells, or a combination thereof. In some embodiments, the cellular system comprises prokaryotic cells, eukaryotic cells, and combinations thereof. In some embodiments, the cellular system comprises *in vitro* expression of proteins based on cellular components, such as ribosomes.

Bacterial cells of the present disclosure include, without limitation, *Escherichia* spp., *Streptomyces* spp., *Zymomonas* spp., *Acetobacter* spp., *Citrobacter* spp., *Synechocystis* spp., *Rhizobium* spp., *Clostridium* spp., *Corynebacterium* spp., *Streptococcus* spp., *Xanthomonas* spp., *Lactobacillus* spp., *Lactococcus* spp., *Bacillus* spp., *Alcaligenes* spp., *Pseudomonas* spp., *Aeromonas* spp., *Azotobacter* spp., *Comamonas* spp., *Mycobacterium* spp., *Rhodococcus* spp., *Gluconobacter* spp., *Ralstonia* spp., *Acidithiobacillus* spp., *Microtholunatus* spp., *Geobacter* spp., *Geobacillus* spp., *Arthrobacter* spp., *Flavobacterium* spp., *Serratia* spp., *Saccharopolyspora* spp., *Thermus* spp., *Stenotrophomonas* spp., *Chromobacterium* spp., *Sinorhizobium* spp., *Saccharopolyspora* spp., *Agrobacterium* spp., *Pantoea* spp., and *Vibrio natriegens*.

Yeast cells of the present disclosure include, without limitation, engineered *Saccharomyces* spp., *Schizosaccharomyces*, *Hansenula*, *Candida*, *Kluyveromyces*, *Yarrowia*, *Candida boidinii*, and *Pichia*. According to the current disclosure, a yeast as claimed herein are eukaryotic, single-celled microorganisms classified as members of the fungus kingdom. Yeasts are unicellular organisms which evolved from multicellular ancestors but with some species useful for the current disclosure being those that have the ability to develop multicellular characteristics by forming strings of connected budding cells known as pseudo hyphae or false hyphae.

In some embodiments, cell pellets are harvested from the cellular system expressing CYP450. In some embodiments, the cell pellets may be resuspended at various concentrations. In some embodiments, the cell pellets are resuspended at a concentration of 1 g/L to 250 g/L. In some embodiments, the cell pellets harvested from the cellular system are resuspended at a concentration of 1 g/L, 10 g/L, 25 g/L, 50 g/L, 75 g/L, 100 g/L, 125 g/L, 150 g/L, 175 g/L, 200 g/L, 225 g/L, or 250 g/L.

Cell Culture

A cell culture refers to any cell or cells that are in a culture. Culturing is the process in which cells are grown under controlled conditions, typically outside of their natural environment. For example, cells, such as yeast cells, may be grown as a cell suspension in liquid nutrient broth. A cell culture includes, but is not limited to, a bacterial cell culture, a yeast cell culture, a plant cell culture, and an animal cell

culture. In some embodiments, the cell culture comprises bacterial cells, yeast cells, or a combination thereof.

A bacterial cell culture of the present disclosure comprises bacterial cells including, but not limited to, *Escherichia* spp., *Streptomyces* spp., *Zymomonas* spp., *Acetobacter* spp., *Citrobacter* spp., *Synechocystis* spp., *Rhizobium* spp., *Clostridium* spp., *Corynebacterium* spp., *Streptococcus* spp., *Xanthomonas* spp., *Lactobacillus* spp., *Lactococcus* spp., *Bacillus* spp., *Alcaligenes* spp., *Pseudomonas* spp., *Aeromonas* spp., *Azotobacter* spp., *Comamonas* spp., *Mycobacterium* spp., *Rhodococcus* spp., *Gluconobacter* spp., *Ralstonia* spp., *Acidithiobacillus* spp., *Microtholunatus* spp., *Geobacter* spp., *Geobacillus* spp., *Arthrobacter* spp., *Flavobacterium* spp., *Serratia* spp., *Saccharopolyspora* spp., *Thermus* spp., *Stenotrophomonas* spp., *Chromobacterium* spp., *Sinorhizobium* spp., *Saccharopolyspora* spp., *Agrobacterium* spp., *Pantoea* spp., and *Vibrio natriegens*.

A yeast cell culture of the present disclosure comprises yeast cells including, but not limited to *Saccharomyces* spp., *Schizosaccharomyces*, *Hansenula*, *Candida*, *Kluyveromyces*, *Yarrowia*, *Candida boidinii*, and *Pichia*.

In some embodiments, a cell culture as described herein is contacted with hydroxy fatty acids as described herein. In some embodiments, a cell culture as described herein is contacted with hydroxy fatty acids as described herein and corn steep liquor, e.g., at a ratio of 1:1 hydroxy fatty acids to corn steep liquor.

In some embodiments, cells are cultured at a temperature of 16° C. to 40° C. For example, cells may be cultured at a temperature of 16° C., 17° C., 18° C., 19° C., 20° C., 21° C., 22° C., 23° C., 24° C., 25° C., 26° C., 27° C., 28° C., 29° C., 30° C., 31° C., 32° C., 33° C., 34° C., 35° C., 36° C., 37° C., 38° C., 39° C. or 40° C.

In some embodiments, cells are cultured for a period of time of 0.5 hours to 96 hours, or more. For example, cells may be cultured for a period of time of 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, or 72 hours. Typically, cells, such as bacterial cells, are cultured for a period of time of 12 to 24 hours. In some embodiments, cells are cultured for 12 to 24 hours at a temperature of 37° C. In some embodiments, cells are cultured for 12 to 24 hours at a temperature of 16° C.

In some embodiments, cells are cultured to a density of 1×10^8 (OD₆₀₀<1) to 2×10^{11} (OD~200) viable cells/ml cell culture medium. In some embodiments, cells are cultured to a density of 1×10^8 , 2×10^8 , 3×10^8 , 4×10^8 , 5×10^8 , 6×10^8 , 7×10^8 , 8×10^8 , 9×10^8 , 1×10^9 , 2×10^9 , 3×10^9 , 4×10^9 , 5×10^9 , 6×10^9 , 7×10^9 , 8×10^9 , 9×10^9 , 1×10^{10} , 2×10^{10} , 3×10^{10} , 4×10^{10} , 5×10^{10} , 6×10^{10} , 7×10^{10} , 8×10^{10} , 9×10^{10} , 1×10^{11} , or 2×10^{11} viable cells/ml. (Conversion factor: OD 1=8×10⁸ cells/ml).

Lactones

Lactones are cyclic carboxylic esters containing a 1-oxa-cycloalkane-2-one structure (—(C=O)—O—) or analogues formed from intramolecular esterification. The present invention relates to lactones (for example, γ-lactones and δ-lactones) formed from straight chain hydroxy carboxylic acids in which the hydroxyl group has reacted with the acid group to form a cyclic carboxylic ester.

The International Union of Pure and Applied Chemistry (IUPAC) names of lactones are derived by adding the term "lactone" at the end of the name of the parent acid. The common names of lactones, used more often than IUPAC names, are formed by changing the -ic acid ending by the hydroxy acid to -olactone. Greek letter designates the carbon atom that bears the hydroxy group to close the ring in the parent hydroxyl carboxylic acid. In other words, Greek letter prefix specifies the number of carbon atoms in the hetero-

cycle—that is, the distance between the relevant —OH and the —COOH groups along said backbone of the parent hydroxyl carboxylic acid. The first carbon atom after the carbon in the —COOH group on the parent compound is labelled α , the second will be labeled β , and so forth. Therefore, the Greek letter prefixes also indicate the size of the lactone ring: α -lactone=3-membered ring, β -lactone=4-membered ring, γ -lactone=5-membered ring; and δ -lactone=6-membered ring.

To illustrate, the lactone derived from the cyclization of γ -hydroxybutyric acid is known as 4-hydroxybutanoic acid lactone or γ -butyrolactone. Similarly, 5-hydroxypentanoic acid is referred as δ -hydroxypentanoic acid and the corresponding lactone is known as 5-hydroxypentanoic acid lactone or δ -valerolactone.

Hydroxy Straight Chain Fatty Acids and Subterminal Hydroxy Groups

Straight chain fatty acid consists of a straight hydrocarbon chain that includes, at the end, a carboxyl group (—COOH). It is the presence of this carboxylic acid group that makes this chemical entity an acid. The straight chain fatty acids may optionally have unsaturated bond between the carbon atom. Branched chain fatty acids are usually saturated fatty acids with one or more alkyl branches substituted on the carbon chain.

One or more hydrogen atoms attached to the hydrocarbon back bone of the straight chain fatty acid may also be replaced by a hydroxyl (—OH) group. Straight chain carboxylic acid with one or more hydroxyl groups is referred as a hydroxy carboxylic acid. Ricinoleic acid is a naturally occurring hydroxy carboxylic acid.

The present invention is related to the replacement or substitution of a hydrogen atom in a saturated straight chain fatty acid with a hydroxyl group using a recombinant microbial organism expressing recombinant protein with the hydroxylase enzyme activity. In a preferred embodiment, substitution of the hydroxyl group takes place at one of the —CH₂— groups closest to the terminal methyl end group of the straight chain fatty acid. In general terminology, n or ω refers to the methyl end group of the straight chain fatty acid while the number n - x of ω - x refers to the location of the carbon atom on the back bone where x indicates the number of carbons away from the methyl end group. To illustrate, in the example of stearic acid (or octadecanoic acid), a C18:0 straight chain fatty acid with the formula CH₃(CH₂)₁₆COOH, ω -3 would mean the fifteenth carbon in the back bone which is three carbon removed from the methyl end group. Similarly, in the same example, ω -1 would mean the seventeenth carbon in the back bone which is one carbon removed from the methyl end group.

As used in the present invention, when a hydroxyl group is introduced into a straight chain fatty acid specifically at positions ω -1, ω -2 and ω -3 using the recombinant DNA technology according to the present invention, the resulting hydroxylated straight chain fatty acid molecule is referred to as a subterminal hydroxy fatty acid, and the hydroxyl group at positions ω -1, ω -2 and ω -3 in such fatty acid molecule are referred as a subterminal hydroxyl groups.

L1

Microbial Culture Media

The microbial organisms useful in the present invention including wild type and recombinant bacterial, yeast and fungal strains are grown in specified culture medium. The microbial culture medium may be minimal growth medium or rich growth medium. The minimal growth medium comprises a fortified mixture of inorganic compounds and one or more organic nutrients such as glucose, sucrose, or glycerol.

The rich growth medium comprises yeast extract, peptone and organic nutrient such as dextrose, sucrose or glycerol. In preferred embodiments, the rich growth medium is supplemented with corn steep liquor to enhance the growth of the microbial organisms.

Synthetic Biology

Standard recombinant DNA and molecular cloning techniques used here are well known in the art and are described, for example, by Sambrook, J., Fritsch, E. F. and Maniatis, T. MOLECULAR CLONING: A LABORATORY MANUAL, 2nd ed.; Cold Spring Harbor Laboratory: Cold Spring Harbor, N.Y., 1989 (hereinafter “Maniatis”); and by Silhavy, T. J., Bannan, M. L. and Enquist, L. W. EXPERIMENTS WITH GENE FUSIONS; Cold Spring Harbor Laboratory: Cold Spring Harbor, N.Y., 1984; and Ausubel, F. M. et al., IN CURRENT PROTOCOLS IN MOLECULAR BIOLOGY, published by GREENE PUBLISHING AND WILEY-INTERSCIENCE, 1987; (the entirety of each of which is hereby incorporated herein by reference).

Bacterial Production Systems

Expression of proteins in prokaryotes is most often carried out in a bacterial host cell with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion proteins. Fusion vectors add a number of amino acids to a protein encoded therein, usually to the amino terminus of the recombinant protein. Such fusion vectors typically serve three purposes: 1) to increase expression of recombinant protein; 2) to increase the solubility of the recombinant protein; and 3) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification. Often, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such vectors are within the scope of the present disclosure.

In an embodiment, the expression vector includes those genetic elements for expression of the recombinant polypeptide in bacterial cells. The elements for transcription and translation in the bacterial cell can include a promoter, a coding region for the protein complex, and a transcriptional terminator.

A person of ordinary skill in the art will be aware of the molecular biology techniques available for the preparation of expression vectors. The polynucleotide used for incorporation into the expression vector of the subject technology, as described above, can be prepared by routine techniques such as polymerase chain reaction (PCR).

A number of molecular biology techniques have been developed to operably link DNA to vectors via complementary cohesive termini. In one embodiment, complementary homopolymer tracts can be added to the nucleic acid molecule to be inserted into the vector DNA. The vector and nucleic acid molecule are then joined by hydrogen bonding between the complementary homopolymeric tails to form recombinant DNA molecules.

In an alternative embodiment, synthetic linkers containing one or more restriction sites are used to operably link the polynucleotide of the subject technology to the expression vector. In an embodiment, the polynucleotide is generated by restriction endonuclease digestion. In an embodiment, the nucleic acid molecule is treated with bacteriophage T4 DNA polymerase or *E. coli* DNA polymerase I, enzymes that remove protruding, 3'-single-stranded termini with their 3'-5'-exonucleolytic activities and fill-in recessed 3'-ends with their polymerizing activities, thereby generating blunt-ended DNA segments. The blunt-ended segments are then incubated with a large molar excess of linker molecules in

the presence of an enzyme that is able to catalyze the ligation of blunt-ended DNA molecules, such as bacteriophage T4 DNA ligase. Thus, the product of the reaction is a polynucleotide carrying polymeric linker sequences at its ends. These polynucleotides are then cleaved with the appropriate restriction enzyme and ligated to an expression vector that has been cleaved with an enzyme that produces termini compatible with those of the polynucleotide.

Alternatively, a vector having ligation-independent cloning (LIC) sites can be employed. The required PCR amplified polynucleotide can then be cloned into the LIC vector without restriction digest or ligation (Aslanidis and de Jong, NUCL. ACID. RES. 18 6069-74, (1990), Haun, et al, BIOTECHNIQUES 13, 515-18 (1992), which is incorporated herein by reference to the extent it is consistent herewith).

In an embodiment, in order to isolate and/or modify the polynucleotide of interest for insertion into the chosen plasmid, it is suitable to use PCR. Appropriate primers for use in PCR preparation of the sequence can be designed to isolate the required coding region of the nucleic acid molecule, to add restriction endonuclease or LIC sites and to place the coding region in the desired reading frame.

In an embodiment, a polynucleotide for incorporation into an expression vector of the subject technology is prepared by the use of PCR using appropriate oligonucleotide primers. The coding region is amplified, whilst the primers themselves become incorporated into the amplified sequence product. In an embodiment, the amplification primers contain restriction endonuclease recognition sites, which allow the amplified sequence product to be cloned into an appropriate vector.

The expression vectors can be introduced into plant or microbial host cells by conventional transformation or transfection techniques. Transformation of appropriate cells with an expression vector of the subject technology is accomplished by methods known in the art and typically depends on both the type of vector and cell. Suitable techniques include calcium phosphate or calcium chloride co-precipitation, DEAE-dextran mediated transfection, lipofection, chemoporation or electroporation.

Successfully transformed cells, that is, those cells containing the expression vector, can be identified by techniques well known in the art. For example, cells transfected with an expression vector of the subject technology can be cultured to produce polypeptides described herein. Cells can be examined for the presence of the expression vector DNA by techniques well known in the art.

The host cells can contain a single copy of the expression vector described previously, or alternatively, multiple copies of the expression vector.

In some embodiments, the transformed cell is an animal cell, an insect cell, a plant cell, an algal cell, a fungal cell, or a yeast cell. In some embodiments, the cell is a plant cell selected from the group consisting of: canola plant cell, a rapeseed plant cell, a palm plant cell, a sunflower plant cell, a cotton plant cell, a corn plant cell, a peanut plant cell, a flax plant cell, a sesame plant cell, a soybean plant cell, and a *petunia* plant cell.

Microbial host cell expression systems and expression vectors containing regulatory sequences that direct high-level expression of foreign proteins are well known to those skilled in the art. Any of these could be used to construct vectors for expression of the recombinant polypeptide of the subject technology in a microbial host cell. These vectors could then be introduced into appropriate microorganisms via transformation to allow for high level expression of the recombinant polypeptide of the subject technology.

Vectors or cassettes useful for the transformation of suitable microbial host cells are well known in the art. Typically the vector or cassette contains sequences directing transcription and translation of the relevant polynucleotide, a selectable marker, and sequences allowing autonomous replication or chromosomal integration. Suitable vectors comprise a region 5' of the polynucleotide which harbors transcriptional initiation controls and a region 3' of the DNA fragment which controls transcriptional termination. It is preferred for both control regions to be derived from genes homologous to the transformed host cell, although it is to be understood that such control regions need not be derived from the genes native to the specific species chosen as a host.

Initiation control regions or promoters, which are useful to drive expression of the recombinant polypeptide in the desired microbial host cell are numerous and familiar to those skilled in the art. Virtually any promoter capable of driving these genes is suitable for the subject technology including but not limited to CYC1, HIS3, GAL1, GAL10, ADHI, PGK, PH05, GAPDH, ADC1, TRP1, URA3, LEU2, ENO, TPI (useful for expression in *Saccharomyces*); AOX1 (useful for expression in *Pichia*); and lac, trp, JPL, IPR, T7, tac, and trc (useful for expression in *Escherichia coli*).

Termination control regions may also be derived from various genes native to the microbial hosts. A termination site optionally may be included for the microbial hosts described herein.

In plant cells, the expression vectors of the subject technology can include a coding region operably linked to promoters capable of directing expression of the recombinant polypeptide of the subject technology in the desired tissues at the desired stage of development. For reasons of convenience, the polynucleotides to be expressed may comprise promoter sequences and translation leader sequences derived from the same polynucleotide. 3' non-coding sequences encoding transcription termination signals should also be present. The expression vectors may also comprise one or more introns in order to facilitate polynucleotide expression.

For plant host cells, any combination of any promoter and any terminator capable of inducing expression of a coding region may be used in the vector sequences of the subject technology. Some suitable examples of promoters and terminators include those from nopaline synthase (nos), octopine synthase (ocs) and cauliflower mosaic virus (CaMV) genes. One type of efficient plant promoter that may be used is a high-level plant promoter. Such promoters, in operable linkage with an expression vector of the subject technology should be capable of promoting the expression of the vector. High level plant promoters that may be used in the subject technology include the promoter of the small subunit(s) of the ribulose-1,5-bisphosphate carboxylase for example from soybean (Berry-Lowe et al., J. MOLECULAR AND APP. GEN., 1:483-98 (1982), the entirety of which is hereby incorporated herein to the extent it is consistent herewith), and the promoter of the chlorophyll binding protein. These two promoters are known to be light-induced in plant cells (see, for example, GENETIC ENGINEERING OF PLANTS, AN AGRICULTURAL PERSPECTIVE, A. Cashmore, Plenum, N.Y. (1983), pages 29-38; Coruzzi, G. et al., THE JOURNAL OF BIOLOGICAL CHEMISTRY, 258:1399 (1983), and Dunsmuir, P. et al., JOURNAL OF MOLECULAR AND APPLIED GENETICS, 2:285 (1983), each of which is hereby incorporated herein by reference to the extent they are consistent herewith).

Analysis of Sequence Similarity Using Identity Scoring

As used herein "sequence identity" refers to the extent to which two optimally aligned polynucleotide or peptide

sequences are invariant throughout a window of alignment of components, e.g., nucleotides or amino acids. An "identity fraction" for aligned segments of a test sequence and a reference sequence is the number of identical components which are shared by the two aligned sequences divided by the total number of components in reference sequence segment, i.e., the entire reference sequence or a smaller defined part of the reference sequence.

As used herein, the term "percent sequence identity" or "percent identity" refers to the percentage of identical nucleotides in a linear polynucleotide sequence of a reference ("query") polynucleotide molecule (or its complementary strand) as compared to a test ("subject") polynucleotide molecule (or its complementary strand) when the two sequences are optimally aligned (with appropriate nucleotide insertions, deletions, or gaps totaling less than 20 percent of the reference sequence over the window of comparison). Optimal alignment of sequences for aligning a comparison window are well known to those skilled in the art and may be conducted by tools such as the local homology algorithm of Smith and Waterman, the homology alignment algorithm of Needleman and Wunsch, the search for similarity method of Pearson and Lipman, and preferably by computerized implementations of these algorithms such as GAP, BESTFIT, FASTA, and TFASTA available as part of the GCG® Wisconsin Package® (Accelrys Inc., Burlington, MA). An "identity fraction" for aligned segments of a test sequence and a reference sequence is the number of identical components which are shared by the two aligned sequences divided by the total number of components in the reference sequence segment, i.e., the entire reference sequence or a smaller defined part of the reference sequence. Percent sequence identity is represented as the identity fraction multiplied by 100. The comparison of one or more polynucleotide sequences may be to a full-length polynucleotide sequence or a portion thereof, or to a longer polynucleotide sequence. For purposes of this disclosure "percent identity" may also be determined using BLASTX version 2.0 for translated nucleotide sequences and BLASTN version 2.0 for polynucleotide sequences.

The percent of sequence identity is preferably determined using the "Best Fit" or "Gap" program of the Sequence Analysis Software Package™ (Version 10; Genetics Computer Group, Inc., Madison, WI). "Gap" utilizes the algorithm of Needleman and Wunsch (Needleman and Wunsch, *JOURNAL OF MOLECULAR BIOLOGY* 48:443-53, 1970) to find the alignment of two sequences that maximizes the number of matches and minimizes the number of gaps. "BestFit" performs an optimal alignment of the best segment of similarity between two sequences and inserts gaps to maximize the number of matches using the local homology algorithm of Smith and Waterman (Smith and Waterman, *ADVANCES IN APPLIED MATHEMATICS*, 2:482-489, 1981, Smith et al., *NUCLEIC ACIDS RESEARCH* 11:2205-2220, 1983). The percent identity is most preferably determined using the "Best Fit" program.

Useful methods for determining sequence identity are also disclosed in the Basic Local Alignment Search Tool (BLAST) programs which are publicly available from National Center Biotechnology Information (NCBI) at the National Library of Medicine, National Institute of Health, Bethesda, Md. 20894; see BLAST Manual, Altschul et al., NCBI, NLM, NIH; Altschul et al., *J. MOL. BIOL.* 215:403-10 (1990); version 2.0 or higher of BLAST programs allows the introduction of gaps (deletions and insertions) into alignments; for peptide sequence BLASTX can be used to

determine sequence identity; and, for polynucleotide sequence BLASTN can be used to determine sequence identity.

As used herein, the term "substantial percent sequence identity" refers to a percent sequence identity of at least about 70% sequence identity, at least about 80% sequence identity, at least about 85% identity, at least about 90% sequence identity, or even greater sequence identity, such as about 98% or about 99% sequence identity. Thus, one embodiment of the disclosure is a polynucleotide molecule that has at least about 70% sequence identity, at least about 80% sequence identity, at least about 85% identity, at least about 90% sequence identity, or even greater sequence identity, such as about 98% or about 99% sequence identity with a polynucleotide sequence described herein. Polynucleotide molecules that have the activity of the Cytochrome P450 genes of the current disclosure are capable of directing the production of a variety of γ - and delta-lactones. Such polynucleotide molecules can have a substantial percent sequence identity to the polynucleotide sequences provided herein and are encompassed within the scope of this disclosure.

Identity and Similarity

Identity is the fraction of amino acids that are the same between a pair of sequences after an alignment of the sequences (which can be done using only sequence information or structural information or some other information, but usually it is based on sequence information alone), and similarity is the score assigned based on an alignment using some similarity matrix. The similarity index can be any one of the following BLOSUM62, PAM250, or GONNET, or any matrix used by one skilled in the art for the sequence alignment of proteins.

Identity is the degree of correspondence between two sub-sequences (no gaps between the sequences). An identity of 25% or higher implies similarity of function, while 18-25% implies similarity of structure or function. Keep in mind that two completely unrelated or random sequences (that are greater than 100 residues) can have higher than 20% identity. Similarity is the degree of resemblance between two sequences when they are compared. This is dependent on their identity.

Explanation of Terms Used Herein

"Coding sequence" is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to a DNA sequence that encodes for a specific amino acid sequence.

"Protein Expression". Protein production can occur after gene expression. It consists of the stages after DNA has been transcribed to messenger RNA (mRNA). The mRNA is then translated into polypeptide chains, which are ultimately folded into proteins. DNA is present in the cells through transfection—a process of deliberately introducing nucleic acids into cells. The term is often used for non-viral methods in eukaryotic cells. It may also refer to other methods and cell types, although other terms are preferred: "transformation" is more often used to describe non-viral DNA transfer in bacteria, non-animal eukaryotic cells, including plant cells. In animal cells, transfection is the preferred term as transformation is also used to refer to progression to a cancerous state (carcinogenesis) in these cells. Transduction is often used to describe virus-mediated DNA

transfer. Transformation, transduction, and viral infection are included under the definition of transfection for this application.

As used herein, the singular forms “a, an” and “the” include plural references unless the content clearly dictates otherwise.

To the extent that the term “include,” “have,” or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

The term “complementary” is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is used without limitation to describe the relationship between nucleotide bases that are capable to hybridizing to one another. For example, with respect to DNA, adenosine is complementary to thymine and cytosine is complementary to guanine. Accordingly, the subjection technology also includes isolated nucleic acid fragments that are complementary to the complete sequences as reported in the accompanying Sequence Listing as well as those substantially similar nucleic acid sequences.

The terms “nucleic acid” and “nucleotide” are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art and are used without limitation to refer to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogues of natural nucleotides that have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally-occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified or degenerate variants thereof (e.g., degenerate codon substitutions) and complementary sequences, as well as the sequence explicitly indicated.

The term “isolated” is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and when used in the context of an isolated nucleic acid or an isolated polypeptide, is used without limitation to refer to a nucleic acid or polypeptide that, by the hand of man, exists apart from its native environment and is therefore not a product of nature. An isolated nucleic acid or polypeptide can exist in a purified form or can exist in a non-native environment such as, for example, in a transgenic host cell.

The term “degenerate variant” refers to a nucleic acid sequence having a residue sequence that differs from a reference nucleic acid sequence by one or more degenerate codon substitutions. Degenerate codon substitutions can be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed base and/or deoxyinosine residues. A nucleic acid sequence and all of its degenerate variants will express the same amino acid or polypeptide.

The terms “polypeptide,” “protein,” and “peptide” are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art; the three terms are sometimes used interchangeably, and are used without limitation to refer to a polymer of amino acids, or amino acid analogs, regardless of its size or function. Although “protein” is often used in reference to relatively large polypep-

tides, and “peptide” is often used in reference to small polypeptides, usage of these terms in the art overlaps and varies. The term “polypeptide” as used herein refers to peptides, polypeptides, and proteins, unless otherwise noted. The terms “protein,” “polypeptide,” and “peptide” are used interchangeably herein when referring to a polynucleotide product. Thus, exemplary polypeptides include polynucleotide products, naturally occurring proteins, homologs, orthologs, paralog, fragments and other equivalents, variants, and analogs of the foregoing.

The terms “polypeptide fragment” and “fragment,” when used in reference to a reference polypeptide, are to be given their ordinary and customary meanings to a person of ordinary skill in the art and are used without limitation to refer to a polypeptide in which amino acid residues are deleted as compared to the reference polypeptide itself, but where the remaining amino acid sequence is usually identical to the corresponding positions in the reference polypeptide. Such deletions can occur at the amino-terminus or carboxy-terminus of the reference polypeptide, or alternatively both.

The term “functional fragment” of a polypeptide or protein refers to a peptide fragment that is a portion of the full-length polypeptide or protein, and has substantially the same biological activity, or carries out substantially the same function as the full-length polypeptide or protein (e.g., carrying out the same enzymatic reaction).

The terms “variant polypeptide,” “modified amino acid sequence” or “modified polypeptide,” which are used interchangeably, refer to an amino acid sequence that is different from the reference polypeptide by one or more amino acids, e.g., by one or more amino acid substitutions, deletions, and/or additions. In an aspect, a variant is a “functional variant” which retains some or all of the ability of the reference polypeptide.

The term “functional variant” further includes conservatively substituted variants. The term “conservatively substituted variant” refers to a peptide having an amino acid sequence that differs from a reference peptide by one or more conservative amino acid substitutions and maintains some or all of the activity of the reference peptide. A “conservative amino acid substitution” is a substitution of an amino acid residue with a functionally similar residue. Examples of conservative substitutions include the substitution of one non-polar (hydrophobic) residue such as isoleucine, valine, leucine or methionine for another; the substitution of one charged or polar (hydrophilic) residue for another such as between arginine and lysine, between glutamine and asparagine, between threonine and serine; the substitution of one basic residue such as lysine or arginine for another; or the substitution of one acidic residue, such as aspartic acid or glutamic acid for another; or the substitution of one aromatic residue, such as phenylalanine, tyrosine, or tryptophan for another. Such substitutions are expected to have little or no effect on the apparent molecular weight or isoelectric point of the protein or polypeptide. The phrase “conservatively substituted variant” also includes peptides wherein a residue is replaced with a chemically-derivatized residue, provided that the resulting peptide maintains some or all of the activity of the reference peptide as described herein.

The term “variant,” in connection with the polypeptides of the subject technology, further includes a functionally active polypeptide having an amino acid sequence at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%,

at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, and even 100% identical to the amino acid sequence of a reference polypeptide.

The term "homologous" in all its grammatical forms and spelling variations refers to the relationship between polynucleotides or polypeptides that possess a "common evolutionary origin," including polynucleotides or polypeptides from super families and homologous polynucleotides or proteins from different species (Reeck et al., *CELL* 50:667, 1987). Such polynucleotides or polypeptides have sequence homology, as reflected by their sequence similarity, whether in terms of percent identity or the presence of specific amino acids or motifs at conserved positions. For example, two homologous polypeptides can have amino acid sequences that are at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, and even 100% identical.

"Suitable regulatory sequences" is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to nucleotide sequences located upstream (5' non-coding sequences), within, or downstream (3' non-coding sequences) of a coding sequence, and which influence the transcription, RNA processing or stability, or translation of the associated coding sequence. Regulatory sequences may include promoters, translation leader sequences, introns, and polyadenylation recognition sequences.

"Promoter" is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to a DNA sequence capable of controlling the expression of a coding sequence or functional RNA. In general, a coding sequence is located 3' to a promoter sequence. Promoters may be derived in their entirety from a native gene, or be composed of different elements derived from different promoters found in nature, or even comprise synthetic DNA segments. It is understood by those skilled in the art that different promoters may direct the expression of a gene in different tissues or cell types, or at different stages of development, or in response to different environmental conditions. Promoters, which cause a gene to be expressed in most cell types at most times, are commonly referred to as "constitutive promoters." It is further recognized that since in most cases the exact boundaries of regulatory sequences have not been completely defined, DNA fragments of different lengths may have identical promoter activity.

The term "operably linked" refers to the association of nucleic acid sequences on a single nucleic acid fragment so that the function of one is affected by the other. For example, a promoter is operably linked with a coding sequence when it is capable of affecting the expression of that coding sequence (i.e., that the coding sequence is under the transcriptional control of the promoter). Coding sequences can be operably linked to regulatory sequences in sense or antisense orientation.

The term "expression" as used herein, is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to the transcription and stable accumulation of sense (mRNA) or antisense RNA derived from the nucleic acid fragment of the subject technology. "Over-expression" refers to the produc-

tion of a gene product in transgenic or recombinant organisms that exceeds levels of production in normal or non-transformed organisms.

"Transformation" is to be given its ordinary and customary meaning to a person of reasonable skill in the field, and is used without limitation to refer to the transfer of a polynucleotide into a target cell for further expression by that cell. The transferred polynucleotide can be incorporated into the genome or chromosomal DNA of a target cell, resulting in genetically stable inheritance, or it can replicate independent of the host chromosomal. Host organisms containing the transformed nucleic acid fragments are referred to as "transgenic" or "recombinant" or "transformed" organisms.

The terms "transformed," "transgenic," and "recombinant," when used herein in connection with host cells, are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art, and are used without limitation to refer to a cell of a host organism, such as a plant or microbial cell, into which a heterologous nucleic acid molecule has been introduced. The nucleic acid molecule can be stably integrated into the genome of the host cell, or the nucleic acid molecule can be present as an extrachromosomal molecule. Such an extrachromosomal molecule can be auto-replicating. Transformed cells, tissues, or subjects are understood to encompass not only the end product of a transformation process, but also transgenic progeny thereof.

The terms "recombinant," "heterologous," and "exogenous," when used herein in connection with polynucleotides, are to be given their ordinary and customary meanings to a person of ordinary skill in the art, and are used without limitation to refer to a polynucleotide (e.g., a DNA sequence or a gene) that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous gene in a host cell includes a gene that is endogenous to the particular host cell but has been modified through, for example, the use of site-directed mutagenesis or other recombinant techniques. The terms also include non-naturally occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position or form within the host cell in which the element is not ordinarily found.

Similarly, the terms "recombinant," "heterologous," and "exogenous," when used herein in connection with a polypeptide or amino acid sequence, means a polypeptide or amino acid sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, recombinant DNA segments can be expressed in a host cell to produce a recombinant polypeptide.

The terms "plasmid," "vector," and "cassette" are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art and are used without limitation to refer to an extra chromosomal element often carrying genes which are not part of the central metabolism of the cell, and usually in the form of circular double-stranded DNA molecules. Such elements may be autonomously replicating sequences, genome integrating sequences, phage or nucleotide sequences, linear or circular, of a single- or double-stranded DNA or RNA, derived from any source, in which a number of nucleotide sequences have been joined or recombined into a unique construction which is capable of introducing a promoter fragment and DNA sequence for a selected gene product along with appropriate

3' untranslated sequence into a cell. "Transformation cassette" refers to a specific vector containing a foreign gene and having elements in addition to the foreign gene that facilitate transformation of a particular host cell. "Expression cassette" refers to a specific vector containing a foreign gene and having elements in addition to the foreign gene that allow for enhanced expression of that gene in a foreign host.

The term "cellular system" as used herein refers to any cells that provide for the expression of ectopic proteins. In some embodiments, the cellular system comprises bacterial cells, yeast cells, plant cells, and animal cells. In some embodiments, the cellular system, comprises prokaryotic cells and/or eukaryotic cells. In some embodiments, the cellular system comprises in vitro expression of proteins based on cellular components, such as ribosomes.

The term "cell culture" as used herein refers to a composition comprising cells. In some embodiments, the cell culture is a liquid (e.g., a broth culture). In some embodiments, the cell culture is a solid (e.g., a stab culture). In some embodiments, the cell culture comprises bacterial cells, yeast cells, plant cells, and animal cells. In some embodiments, the cell culture comprises prokaryotic cells and/or eukaryotic cells. In some embodiments, the cell culture comprises in vitro expression of proteins based on cellular components, such as ribosomes.

The terms "incubating" and "incubation" as used herein refers to a process of mixing two or more chemical or biological entities (such as a chemical compound and an enzyme) and allowing them to interact under conditions favorable for producing a new chemical entity such as hydroxy fatty acid (e.g., a subterminal hydroxy fatty acid) as described herein or a lactone (e.g., a gamma-lactone, a delta-lactone, or combinations thereof) as described herein.

"Percent (%) amino acid sequence identity" with respect to the variant polypeptide sequences of the subject technology refers to the percentage of amino acid residues in a candidate sequence that are identical with the amino acid residues of a reference polypeptide, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity.

Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN, ALIGN-2 or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full-length of the sequences being compared. For example, the % amino acid sequence identity may be determined using the sequence comparison program NCBI-BLAST2. The NCBI-BLAST2 sequence comparison program may be downloaded from ncbi.nlm.nih.gov. NCBI-BLAST2 uses several search parameters, wherein all of those search parameters are set to default values including, for example, unmask yes, strand=all, expected occurrences 10, minimum low complexity length=15/5, multi-pass e-value=0.01, constant for multi-pass=25, dropoff for final gapped alignment=25 and scoring matrix=BLOSUM62. In situations where NCBI-BLAST2 is employed for amino acid sequence comparisons, the % amino acid sequence identity of a given amino acid sequence A to, with, or against a given amino acid sequence B (which can alternatively be phrased as a given amino acid sequence A that has or comprises a certain % amino acid sequence identity to, with, or against a given amino acid

sequence B) is calculated as follows: 100 times the fraction X/Y where X is the number of amino acid residues scored as identical matches by the sequence alignment program NCBI-BLAST2 in that program's alignment of A and B, and where Y is the total number of amino acid residues in B. It will be appreciated that where the length of amino acid sequence A is not equal to the length of amino acid sequence B, the % amino acid sequence identity of A to B will not equal the % amino acid sequence identity of B to A.

In this sense, techniques for determining amino acid sequence "similarity" are well known in the art. In general, "similarity" refers to the exact amino acid to amino acid comparison of two or more polypeptides at the appropriate place, where amino acids are identical or possess similar chemical and/or physical properties such as charge or hydrophobicity. A so-termed "percent similarity" may then be determined between the compared polypeptide sequences. Techniques for determining nucleic acid and amino acid sequence identity also are well known in the art and include determining the nucleotide sequence of the mRNA for that gene (usually via a cDNA intermediate) and determining the amino acid sequence encoded therein, and comparing this to a second amino acid sequence. In general, "identity" refers to an exact nucleotide to nucleotide or amino acid to amino acid correspondence of two polynucleotides or polypeptide sequences, respectively. Two or more polynucleotide sequences can be compared by determining their "percent identity", as can two or more amino acid sequences. The programs available in the Wisconsin Sequence Analysis Package, Version 8 (available from Genetics Computer Group, Madison, Wis.), for example, the GAP program, are capable of calculating both the identity between two polynucleotides and the identity and similarity between two polypeptide sequences, respectively. Other programs for calculating identity or similarity between sequences are known by those skilled in the art.

An amino acid position "corresponding to" a reference position refers to a position that aligns with a reference sequence, as identified by aligning the amino acid sequences. Such alignments can be done by hand or by using well-known sequence alignment programs such as ClustalW2, Blast 2, etc.

Unless specified otherwise, the percent identity of two polypeptide or polynucleotide sequences refers to the percentage of identical amino acid residues or nucleotides across the entire length of the shorter of the two sequences.

"Transformation" is used according to its ordinary and customary meaning as understood by a person of ordinary skill in the art, and is used without limitation to refer to the transfer of a polynucleotide into a target cell. The transferred polynucleotide can be incorporated into the genome or chromosomal DNA of a target cell, resulting in genetically stable inheritance, or it can replicate independent of the host chromosomal. Host organisms containing the transformed nucleic acid fragments are referred to as "transgenic" or "recombinant" or "transformed" organisms.

The terms "recombinant," "heterologous," and "exogenous," when used herein in connection with polynucleotides, are used according to their ordinary and customary meanings as understood by a person of ordinary skill in the art, and are used without limitation to refer to a polynucleotide (e.g., a DNA sequence or a gene) that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous gene in a host cell includes a gene that is endogenous to the particular host cell but has been modified through, for example, the use of site-directed mutagenesis or

other recombinant techniques. The terms also include non-naturally occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position or form within the host cell in which the element is not ordinarily found.

Similarly, the terms “recombinant,” “heterologous,” and “exogenous,” when used herein in connection with a polypeptide or amino acid sequence, means a polypeptide or amino acid sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, recombinant DNA segments can be expressed in a host cell to produce a recombinant polypeptide.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs. Although any methods and materials similar to or equivalent to those described herein can be used in the practice or testing of the present disclosure, the preferred materials and methods are described below.

The disclosure will be more fully understood upon consideration of the following non-limiting Examples. It should be understood that these Examples, while indicating preferred embodiments of the subject technology, are given by way of illustration only. From the above discussion and these Examples, one skilled in the art can ascertain the essential characteristics of the subject technology, and without departing from the spirit and scope thereof, can make various changes and modifications of the subject technology to adapt it to various uses and conditions.

All of the features disclosed in this specification may be combined in any combination. Each feature disclosed in this specification may be replaced by an alternative feature serving the same, equivalent, or similar purpose. Thus, unless expressly stated otherwise, each feature disclosed is only an example of a generic series of equivalent or similar features. From the above description, one skilled in the art can easily ascertain the essential characteristics of the present disclosure, and without departing from the spirit and scope thereof, can make various changes and modifications of the present disclosure to adapt it to various usages and conditions. Thus, other embodiments are also within the claims.

EQUIVALENTS AND SCOPE

Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation many equivalents to the specific embodiments described herein. The scope of the present embodiments described herein is not intended to be limited to the above Description, but rather is as set forth in the appended claims. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present disclosure, as defined in the following claims.

The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally

be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03. It should be appreciated that embodiments described in this document using an open-ended transitional phrase (e.g., “comprising”) are also contemplated, in alternative embodiments, as “consisting of” and “consisting

essentially of” the feature described by the open-ended transitional phrase. For example, if the disclosure describes “a composition comprising A and B”, the disclosure also contemplates the alternative embodiments “a composition consisting of A and B” and “a composition consisting essentially of A and B”.

Furthermore, the present disclosure encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, and descriptive terms from one or more of the listed claims is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should be understood that, in general, where the present disclosure, or aspects of the present disclosure, is/are referred to as comprising particular elements and/or features, certain embodiments of the present disclosure or aspects of the present disclosure consist, or consist essentially of, such elements and/or features. For purposes of simplicity, those embodiments have not been specifically set forth in haec verba herein. It is also noted that the terms “comprising” and “containing” are intended to be open and permits the inclusion of additional elements or steps. Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or sub-range within the stated ranges in different embodiments of the present disclosure, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

This application refers to various journal articles, and other publications, all of which are incorporated herein by reference. All references, patents and patent applications disclosed herein are incorporated by reference with respect to the subject matter for which each is cited, which in some cases may encompass the entirety of the document. If there is a conflict between any of the incorporated references and the instant specification, the specification shall control. In addition, any particular embodiment of the present disclosure that falls within the prior art may be explicitly excluded from any one or more of the claims. Because such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment of the present disclosure can be excluded from any claim, for any reason, whether or not related to the existence of prior art.

EXAMPLES

Example 1

Cloning of *M. thermophile* CYP505A30 and *E. coli* Icd

The amino acid sequence of a BM3 homolog of *Myceliophthora thermophile* (CYP505A30) was obtained from UniProt (www.uniprot.org/uniprot/G2QDZ3.fasta) and the corresponding gene was codon optimized for expression in *Escherichia coli* and synthesized by GenScript (Piscataway, NJ). The nucleotide sequence of CYP505A30 from *M. thermophile* with codon-optimization for *E. coli* expression is provided as SEQ ID NO: 1. The amino acid sequence of CYP505A30 is provided as SEQ ID NO: 3.

The resulting gene product was cloned into pETDuet-1 vector (AMP^r, Novagen) through NdeI and XhoI sites. The construct was transformed into BL21 (DE3) cells for expression. To enhance CYP505A30 activity in *E. coli*, an isocitrate dehydrogenase (Icd)-mediated NADPH regenerating system was coupled with CYP505A30. Primers ICD-F, 5'-ggaattcCATatgGAAAGTAAAGTAGTTGT (SEQ ID NO: 7) and ICD-R, 5'-ccgCTCGAGttaCATGTTTCGATGATCGCGT (SEQ ID NO: 8) were used to amplify the icd gene by PCR from the genome of *E. coli* K-12 substr. MG1655. The nucleotide sequence of icd from *E. coli* K-12 substr. MG1655 is provided as SEQ ID NO: 2. The corresponding amino acid sequence is provided as SEQ ID NO: 4.

The resulting PCR product was subcloned into pCDF-Duet-1 vector (Spect^r, Novagen) through NdeI and XhoI sites. The construct was used to transform BL21 (DE3) cells overexpressing CYP505A30 and the transformants were selected on LB (AMP^rSpect^r) plates.

The resulting vectors are used to overexpress *M. thermophile* CYP505A30 and *E. coli* Icd in *E. coli* cells.

Example 2

Production of Gamma- and Delta-Lactones Using *E. coli* Cells Overexpressing *M. Thermophile* CYP505A30 and *E. coli* Icd

This example demonstrates the use of *E. coli* cells overexpressing *M. thermophile* CYP505A30 and *E. coli* Icd in the bioconversion of short chain fatty acids into gamma- and delta-lactones via a subterminal hydroxy fatty acid intermediate.

To produce subterminal (i.e., omega-1 (ω -1), omega-2 (ω -2), and omega-3 (ω -3)) hydroxy fatty acids from corresponding short chain fatty acids, *E. coli* cells overexpressing CYP505A30 and Icd were incubated with medium comprising the short chain fatty acids as substrates.

In a typical experiment, an overnight culture was used to inoculate liquid LB medium (2%) containing 100 mg/L of carbenicillin and 100 mg/L of spectinomycin. The culture was first grown at 37° C. to an OD₆₀₀ of 0.6 and cooled down to 16° C. Then 1 mM IPTG was added to induce protein expression. After 16 h of incubation at 16° C., cells were harvested by centrifugation.

Harvested cell pellets were re-suspended at a concentration of 100 g/L fresh weight in 100 mM potassium phosphate buffer (pH7.0) containing 0.1% Tween 40. Then 3 g/L of lauric acid (C12:0) or 3 g/L of tridecanoic acid (C13:0) together with 10 g/L of isocitric acid trisodium salt were added for biotransformation at 30° C. in a shaker. Samples were taken at different reaction times and analyzed by GC/FID.

GC/FID analysis was conducted on Shimadzu GC-2014 system. The analytical column is Restek RXi-5 ms (thickness 0.25 μ m; length 30 m; diameter 0.25 mm) and the injection temperature is 240° C. under split mode. The temperature gradient is 0-3 min, 100° C.; 3-9 min 100° C.-280° C., a gradient of 30; 9-12 min, 280° C.

As shown in FIG. 2A and FIG. 2B, HO-C12:0 and HO-C13:0 were efficiently produced from the corresponding carboxylic acids C12:0 (dodecanoic acid, also known as lauric acid) and C13:0 (tridecanoic acid, also known as tridecyclic acid) by *E. coli* cells modified according to the procedures described in Example 1. According to Baker et al. (2017), CYP505A30 catalyzed hydroxylation of straight chain fatty acids at the ω -1 to ω -3 positions. However, FIGS.

29

2A-2B show single peaks because GC/FID cannot separate ω -1 to ω -3 hydroxy fatty acids in the absence of derivatization. As a result, the peaks represent a mixture of ω -1 to ω -3 hydroxy fatty acids.

These results evidence production of ω -1 to ω -3 hydroxy fatty acids from corresponding fatty acids by re-suspended *E. coli* cells overexpressing *M. thermophile* CYP505A30 and *E. coli* lcd.

To produce gamma-lactones (5-membered ring) and delta-lactones (6-membered ring) from subterminal hydroxy fatty acids, the mixture of ω -1 to ω -3 hydroxy fatty acids from the previous step was first isolated from the re-suspended *E. coli* cells then inoculated with a *Candida boidinii* culture.

As an example of producing lactones from hydroxy fatty acids, the mixture of HO-C12:0 and the mixture of HO-C13:0, respectively, was mixed with 60 g/L of corn steep liquor (CSL) at a 1:1 ratio, and the resulting medium was autoclaved at 121° C. for 20 min. The media was named as CYP505A30C12 or CYP505A30C13 for C12:0 and C13:0 bioconversion, respectively. Then the medium was inoculated with an overnight yeast culture (e.g., *Candida boidinii*), and the yeast culture was grown in a 30° C. shaker. Samples were taken daily after inoculation and analyzed as described herein.

To analyze γ -lactone production, 0.5 ml *E. coli* culture was taken and 10 μ l 2N HCl was added. The acidified culture was extracted with 0.5 ml ethyl acetate with shaking at room temperature for 60 min. After centrifugation at 14,000 rpm for 15 min, the ethyl acetate phase was used for GC/MS or GC/FID analysis.

GC/MS analysis was conducted on Shimadzu GC-2010 system coupled with GCMS-QP2010S detector. The analytical column is SHRXL-5 MS (thickness 0.25 μ m; length 30 m; diameter 0.25 mm) and the injection temperature is 265° C. under split mode. The temperature gradient is 0-3 min 80° C.; 3-8.7 min 120° C. to 263° C., a gradient of 25; 8.7-10.7 min, 263° C.

As shown in FIG. 3, if the initial starting fatty acid has an even number of carbons such as C12:0 (dodecanoic acid, also known as lauric acid), after β -oxidation in yeast, the predicted products would be δ -hexalactone (DC6) from the ω -1 hydroxy fatty acid intermediate, γ -hexalactone (GC6) from the ω -2 hydroxy fatty acid intermediate, and δ -octalactone (DC8) from the ω -3 hydroxy fatty acid intermediate. This is because in each round of β -oxidation, 2 carbon atoms are removed from the hydroxy fatty acid intermediate. Depending on the position of the OH group in the hydroxy fatty acid, there can be 9, 8, or 7 intervening carbon atoms between the hydroxylated carbon atom and the carboxylated carbon atom in the hydroxy fatty acid. Once β -oxidation has removed enough pairs of carbon atoms such that there are either 2 or 3 intervening carbon atoms remaining, favorable kinetics lead to cyclization into the lactone product (a γ -lactone if there are 2 intervening carbon atoms remaining, and a δ -lactone if there are 3 intervening carbon atoms remaining).

FIG. 4 shows the spectra from GC/MS analysis obtained with CYP505A30C12 (containing ω -1 to ω -3 hydroxy lauric acid) following incubation with *Candida boidinii*. Referring to FIG. 4A, peaks 1, 2, 3, and 4 were analyzed. After comparing the retention times to those of known standards, peak 1 was identified to be the γ -hexalactone GC6, peak 2 was identified to be the δ -hexalactone DC6, and peak 4 was identified to be the δ -octalactone DC8. Peak 3 was identified to be phenethyl alcohol, which is a metabolite of yeast metabolism.

30

As shown in FIG. 5, if the initial starting fatty acid has an odd number of carbons such as C13:0 (tridecanoic acid, also known as tridecylic acid), after β -oxidation in yeast, the predicted products would be γ -valerolactone (GC5) from the ω -1 hydroxy fatty acid intermediate, δ -heptalactone (DC7) from the ω -2 hydroxy fatty acid intermediate, and γ -heptalactone (GC7) from the ω -3 hydroxy fatty acid intermediate. This is because in each round of β -oxidation, 2 carbon atoms are removed from the hydroxy fatty acid intermediate. Depending on the position of the OH group in the hydroxy fatty acid, there can be 10, 9, or 8 intervening carbon atoms between the hydroxylated carbon atom and the carboxylated carbon atom in the hydroxy fatty acid. Once β -oxidation has removed enough pairs of carbon atoms such that there are either 2 or 3 intervening carbon atoms remaining, favorable kinetics lead to cyclization into the lactone product (a γ -lactone if there are 2 intervening carbon atoms remaining, and a δ -lactone if there are 3 intervening carbon atoms remaining).

FIG. 6 shows the spectra from GC/MS analysis obtained with CYP505A30C13 (containing ω -1 to ω -3 hydroxy tridecanoic acid) following incubation with *Candida boidinii*. Referring to FIG. 6A, peaks 1, 2, 3, and 4 were analyzed. After comparing the retention times to those of known standards, peak 2 was identified to be the γ -heptalactone GC7, and peak 3 was identified to be the δ -heptalactone DC7. Peak 1 was identified to be phenethyl alcohol, which is a metabolite of yeast metabolism. Valerolactone (GC5) was not detected by GC/MS most likely due to its low concentration or volatility.

These results demonstrate production of lactones from hydroxy fatty acids by inoculation with a yeast cell culture.

Example 3

Production of Lactones Using *E. coli* Cells Overexpressing *U. Isabellina* Cytochrome P450 Protein

The amino acid sequence of a BM3 homolog from *Umbelopsis isabellina*, a cytochrome P450 protein, was obtained from JGI Genome Portal (www.genome.jgi.doe.gov/cgi-bin/dispGeneModel?db=Umbisal&id=523695) and the corresponding gene was codon-optimized for expression in *Escherichia coli* and synthesized by Gene Universal Inc. (Newark, DE). The nucleotide sequence of BM3 homolog of *Umbelopsis isabellina*, MI2, with codon-optimization for *E. coli* expression is provided below as SEQ ID NO: 5. The amino acid sequence of MI2 is provided below as SEQ ID NO: 6.

The resulting gene product was cloned into pET-32a(+) vector (AMP+, Novagen) through HindIII and XhoI sites. The construct was transformed into BL21 (DE3) cells for expression, and an overnight culture was used to inoculate liquid LB medium (2%) containing 100 mg/L of carbenicillin. After 16 h of incubation at 16° C., cells were harvested by centrifugation.

Harvested cell pellets were resuspended at a concentration of 100 g/L fresh weight in 100 mM potassium phosphate buffer (pH7.0) containing 0.1% Tween 40. Then 1 g/L of lauric acid (C12:0) or 1 g/L of tridecanoic acid (C13:0) was added for biotransformation at 30° C. in a shaker. Samples were taken at different reaction times and the production of hydroxy fatty acids were determined by GC/MS after silylation with MSTFA (N-methyl-N-trimethylsilyl-trifluoroacetamide).

31

FIG. 7A and FIG. 7B demonstrate production of HO-C12:0 and HO-C13:0 from the corresponding carboxylic acids by *E. coli* cells overexpressing BM3 homolog of *Umbelopsis isabellina*, MI2. Although a mass spectrum library search did not find exact matches for peak 1 in the samples, the search indicated that the peak was hydroxy fatty acids.

To produce lactones from the hydroxy fatty acids HO-C12:0 and HO-C13:0, the biotransformation mixture was mixed with 60 g/L of corn steep liquor (CSL) at 1:1 ratio, and the resulting medium was autoclaved at 121° C. for 20 min. The media was labelled as MI2-C12-CSL or MI2-C13-CSL for C12:0 and C13:0 bioconversion, respectively. Then the medium was inoculated with an overnight yeast culture, e.g., *Candida boidinii* and the yeast culture was grown in a 30° C. shaker. Samples were taken daily after inoculation.

To analyze lactone production in yeast cultures, 0.5 ml *Candida boidinii* culture was taken and 10 μ l 2N HCl was added. The acidified culture was extracted with 0.5 ml ethyl acetate with shaking at room temperature for 60 min. After centrifugation at 14,000 rpm for 15 min, the ethyl acetate phase was used for GC/MS or GC/FID analysis.

FIG. 8A and FIG. 8B demonstrate that GC6 is the major lactone derived from C12:0 (via HO-C12:0 as an intermediate) and DC7 is the major lactone derived from C13:0 (via HO-C13:0 as an intermediate), respectively. This suggests that ω -2 hydroxy lauric acid and ω -2 hydroxy tridecanoic acid likely correspond to peak 1 in FIG. 7A and FIG. 7B, respectively, which in turn were derived from C12 and C13, respectively.

Taken together, these results demonstrate that *Umbelopsis isabellina* cytochrome P450 MI2 is an ω -2 fatty acid hydroxylase and can be used for production of γ -hexalactone or 8-heptalactone.

Sequences of Importance

M. thermophile CYP505A30 nucleotide sequence

(SEQ ID NO: 1)
 ATGGCGGATAAGACCACCGAAACCGTGCCGATTCCGGTCCGCCGGGCTG
 CCGCTGGTTGGTAATGCGCTGGCGTTTGATAGCGAACTGCCGCTGCGTACC
 TTCCAGGAATTTGCGGAGGAATACGGCGAGATCTATCGTCTGACCTGCCG
 ACCGGTACCACCTGGTGGTTAGCAGCCAAGCGCTGGTTCACGAACTGTGC
 GACGATAAGCGTTTCAAGAAGCCGGTTGCTGCGGCGCTGGCGGAAGTGCCT
 AACGGCGTTAACGACGGTCTGTTTACC CGCGCTGAAGAGGAGCCGAATGG
 GGCATCGCGCACCGTATTCTGATGCCGGCGTTTGGTCCGGCGAGCATTCAG
 GGC
 ATGTTTACC GAAATGCACGAGATCGCGAGCCAACCTGGCGCTGAAATGGGCG
 CGTCACGGTCCGGACACCCGATTTTCGTTACCGACGATTTTACCCGTCTG
 ACCCTGGATACCTTGGCGCTGTGCACCATGAACTTCCGTTTAAACAGCTAC
 TATCACGACGAACTGCACCCGTTTATCAACGCGATGGGCAACTTTCTGACC
 GAGAGCGGTGCGCGTGCATGCGTCCGGCGATCACCAGCATTTTCCACCAG
 GCGGCGAACCGTAAGTACTGGGAAGATATTGAGGTTCTGCGTAAACCGCG
 CAAGGTGTGCTGGACACCGCTCGTAAGCACCCGACCAACCGTAAAGATCTG
 CTGAGCGCGATGCTGGACGGCGTGGATGCGAAAACCGGTCAGAACTGAGC
 GACAGCAGCATATTGATAACCTGATCACCTTTCTGATTGCGGCCACGAA

32

-continued

ACCACCAGCGGTCTGCTGAGCTTCGCGTTTTACCTGCTGATTAAGCACCAG
 GACGCGTATCGTAAAGCGCAAGAAGAGGTGGATCGTGTTATCGGCAAGGGC
 CCGATTAAAGTTGAACACATCAAGAACTGCCGTACATCGCGCGGTGCTG
 CGTGAAACCTGCGTCTGTGCCCCGACCATTCCGATCATTAAACCGTGCGGCG
 GAAGTTATCGGTGGCAAGTACGCGGTGGCGAAAGATCAGCGTCTGGCGCTG
 CTGCTAAGCAGGACGGCGCAAGCCACCTGGACCCGCGGTTTATGGCGAA
 ACCGCGAAGCAATTATTCGCGAGCGTATGCTGGACGAAAACCTTTGAGCGT
 CTGAACCGTGAGTATCCGGATTGCTGGAAACCGTTCGGTACCGGCATGCGT
 CGCTGCATCGGTCTGCTCCGTTTTCGTGCGCAGGAAGCGGTGCTGGTTATGGCG
 ATGCTGCTGCAAAACTTCGACTTTGTCTGACGATCCGTACTATGAGCTG
 CACTACAAGCAGACCCTGACCACCAAGCCGAAAGACTTCTATATGCGTGCG
 ATCTCTGCGTGATGGCCTGACCGCGACCGAACTGGAGCACCGCTCTGGCGGT
 AACGCGGCGAGCGTGGCGGTAGCGGTGGCGGTGGCGGTGGCGCGAGCAAA
 CCGACCGCGCAGAAAACCGACCCGGCGGAAGCGAAACCGATGAGCATCTTC
 TACGGCAGCAACACCGGTACCTGCGAGAGCGTGGCGCAACGCTCTGGCGACC
 GATGCGGCGAGCCACGGTTATGCTGCGGCGCGGTGGAAACCGTGGACACC
 GCGACCGAGAAGCTGCCGACCGATCGTCCGGTGGTTATCATTACCGCGAGC
 TTCGAGGGTCAGCCCGGACAACGCGCGAAGTTTTCGCGCTGGCTGAAA
 AACCTGGAAGGTGATGAGCTGAAAAACGTGAGCTACGCGGTTTTTCGTTGC
 GGCCACCACGACTGGAGCCAGACCTTTCACCGTATTCGGAAGCTGGTTAC
 CAAACCATGAAAGCGCACGGTGCAGAGCCGATCTGCGACGAAGCGCTGACC
 GATGTGGCGGAGGGTAACATGTTTACCGATTGTTGAACAATGGGAGGACGAT
 GTGTTCTGGCCGGCGGTTCTGTGCGGTTATGGCGCGCGGTGCGGTTGCG
 GAAACCGAGGACGCGCCGGTAGCGATGGTCTGAACATCCACTTTAGCAGC
 CCGCGTAGCAGCACCTGCGTCAGGACGTGCGTGAAGCAGCCGTGGTTGGT
 GAAGCGCTGCTGACCGCGCGGATGCGCGCGCAAGAAACACATTGAAGTT
 CAACTGCCGCGACGCGCGACCTACAAAGTGGGTGATTATCTGGCGGTGCTG
 CCGGTTAACAGCAAGGAGAGCATTTGGTCTGTGTTATCGTAAATTCACGTG
 AGCTGGGACAGCCACGTGACCATCGCGAGCGATCGTTGGACCGCGCTGCCG
 ACCGGTACCCCGGTGCCGCGTACGACGTTCTGGGTAGCTATGTGGAGCTG
 AGCCAAACCGCGACCAACCGTGGTATCTGCGTCTGGCGGATGCGCGGAA
 GATGAGGCGACCAAGGCGGAACCTGCAAAAACCTGGCGGTGATCTGTACACC
 AGCGAGATTAGCTGAAACGTGCGAGCGTTCTGGACCTGCTGGATCGTTTC
 CCGAGCATCAGCTTCCGTTTCGGTACCTTTCTGAGCCTGCTGCCCGGATT
 CGTCCGCGTCAATACAGCATCAGCAGCAGCCCGCTGAACGACCCGAGCCGT
 GCGACCTGACCTATAGCCTGCTGGATAGCCCGAGCCTGGCGAACCCGAGC
 CGTCGTTCTGTTGGCGTTGCGACCAAGCTACCTGAGCAGCCTGGTTCTGTT
 GACAAGCTGCTGGTGGCGTTGCTCCGACCCACACCGCGTTTCTGCTGCCG
 GACGAAGATAAAATGGGTGAAACCGCGATCATTTGCGTGGGTGCGGGTAGC
 GGTCTGGCGCGGTTCCGTTGGTTTTATCCAGGAACGTGCGCGCTGCTGGCG

33

-continued

AAAGGTACCCAACTGGCGGCGCGCTGCTGTTCTACGGTTGCCGTAGCCCCG
 GAGAAGGACGATCTGTATCGTGACGAATTCGATAAATGGCAAGAGAGCGGT
 GCGGTGGATGTTCTGTCGTGCGTTTAGCCGTGTTGATAGCGACGATACCGAG
 GCGCGTGGTTGCCGTCACGTTTACGACCGCTCTGTGGCAGCATCGTGAAGAG
 GTGAAGGCGCTGTGGGACCGTGGCGCGCGTGTGTACGTTTGGCGTAGCCGT
 CAAGTGGGCGAAGGTGTTAAACCCGCGATGGGCCGTATCGTGCTGGGTGAA
 GAGGACGCGGAGGATGCGATCAGCAAGTGGTATGAAACCGTGCCTAATGAC
 CGTTATGCGACCGATGTGTTCTGACTAA

icd gene from *E. coli* K-12 substr. MG1655

(SEQ ID NO: 2)

GAAAGTAAAGTAGTTGTTCCGGCACAAGGCAAGAAGATCACCTGCAAAAC
 GGCAAACTCAACGTTCTGAAATCCGATTATCCCTTACATTGAAGGTGAT
 GGAATCGGTGTAGATGTAACCCAGCCATGCTGAAAGTGGTCGACGCTGCA
 GTCGAGAAAGCCTATAAAGGCGAGCGTAAATCTCCTGGATGGAATTTAC
 ACCGGTGAAAAATCCACACAGGTTTATGGTCAGGACGCTGGCTGCCTGCT
 GAAACTCTTGATCTGATTCTGTAATATCGCGTTGCCATTAAAGGTCCGCTG
 ACCACTCCGGTTGGTGGCGGTATTCTGCTCTCTGAACGTTGCCCTGCGCCAG
 GAACTGGATCTCTACATCTGCTGCGTCCGTACGTTACTATCAGGGCACT
 CCAAGCCCGGTTAAACACCCTGAACGATGCGGTTATCTTCCGTGAA
 AACTCGGAAGACATTTATGCGGGTATCGAATGGAAGCAGACTCTGCCGAC
 GCCGAGAAAGTGATTAATTTCTGCGTGAAGAGATGGGGTGAAGAAAATT
 CGCTTCCCGGAACATTGTGGTATCGGTATTAAGCCGTGTTGGAAGAAGGC
 ACCAAACGTCCTGGTTCGTGACGCGATCGAATACGCAATTGCTAACGATCGT
 GACTCTGTGACTCTGGTGCACAAAGGCAACATCATGAAGTTCACCGAAGGA
 GCGTTTAAAGACTGGGGCTACAGCTGGCGCGTGAAGAGTTTGGCGGTGAA
 CTGATCGACGCTGGCCCGTGGCTGAAAGTTAAAAACCCGAACACTGGCAAA
 GAGATCGTCATTAAAGACGTGATTGCTGATGCATTCCTGCAACAGATCCTG
 CTGCGTCCGGCTGAATATGATGTTATCGCCTGTATGAACCTGAACGGTGAC
 TACATTTCTGACGCCCTGGCAGCGCAGGTTGGCGGTATCGGTATCGCCCT
 GGTGCAACATCGGTGACGAATGCGCCCTGTTTGAAGCCACCCACGGTACT
 GCGCCGAAATATGCCGGTCAGGACAAAGTAAATCCTGGCTCTATTATTCTC
 TCCGCTGAGATGATGCTGCGCCACATGGGTTGGACCGAAGCGGCTGACTTA
 ATTGTTAAAGGTATGGAAGGCGCAATCAACGCGAAAACCGTAACCTATGAC
 TTCGAGCGTCTGATGGATGGCGCTAAACTGCTGAAATGTTTACAGATTTGGT
 GACGCGATCATCGAAAACATGTAA

M. thermophile CYP505A30 amino acid sequence

(SEQ ID NO: 3)

MADKTTETVPIPGPPGLPLVGNALAFDSELPLRTFQEFQAEYGEIYRLTLP
 TGTTLVVSSQALVHELCDKRFKPKPVAAALAEVRNGVNDGLFTAREEPPNW
 GIAHRILMPAFGPASIQGMFTHEMHEIASQLALKWARHGPDPITFVTDFTFR
 LTLDTLALCTMNFNFNSYYHDELHPFINAMGNFLTESGARAMRPAITSIFH
 QAA NRKYWEDIEVLRKTAQGVLDTRRKHPTNRKDLLSAMLGDVDAKTGQKL
 SDSSIIDNLITFLIAGHETTSGLLSFAFYLLIKHQDAYRKAQEEVDVRVIGK

34

-continued

GPIKVEHIKKLPYIAAVLRETLRCLPTIPIINRAAKQDEVIGGKYAVAKDQ
 RLALLLAQSHLDPVYGETAKQFIPERMLDENFERLNREYPCWKPFPGTGM
 RACIGRPFPAWQEAFLVMAMLLQNFDFVLHDPYELHYKQTLTTKPKDFYMR
 AILRDGLTATELEHRLAGNAASVARSGGGGGGPSKPTAQKTSPAEAKPMSI
 FYGSNTGTCSLAQRLATDAASHGYAAAAVEPLDTATEKLPDTPRVVITA
 SFEGQPPDNAAKFCGWLKNLEGDELKNVSYAVFGCGHHDSQTPHRIKPLV
 HQTMAKHAGSPICDEGLTDVAEGNMFTDFEQWEDDVFWPAVRARYGAAGAV
 AETEDAPGSDGLNIHFSSPRSSTLRQDVREATVVGALLTAPDAPPKKHIE
 VQLPDGATYKVDYLAFLPVNSKESIGRVMRKQLSWDSHVTIASDRWTAL
 PTGTPVPAYDVLGSYVELSQPATKRGILRLADAEEATKAELQKLAGDLY
 TSEISLKRASVLDLLDRFPSISLPGTFLSLLPPIRQYSSISSPLNDPS
 RATLTYSLDPSLANPSRRFVGVSATSYLSSLVRGDKLLSVSRPHTHTAFRL
 PDEDKMGETAII CVGAGSGLAPFRGFIQERAAALLAKGTQLAALLFFYGRS
 PEKDDLRYDEFDKWQESGAVDVRRAPSRVDSDDTEARGCRHVQDRLWHDR
 EVKALWDRGARVYVCGSRQVGBGVKTAMGRIVLGEEDAEDAISKWYETVRN
 DRYATDVFD

icd protein from *E. coli* K-12 substr. MG1655

(SEQ ID NO: 4)

MESKVVPVPAQGGKITLQNGKLVNPNPIPIYIEGDIGVDVTPAMLKVDA
 AVEKAYKGERKISWMEIYTGKSTQVYQDVLPAETLDLIREYVAIKGP
 LTTPVGGGIRSLNVALRQELDLYICLRPVRYQGTSPVKHPELTDWVIFR
 ENSEDIYAGIEWKADSADAEEKVIKFLREEMGVKKIRFPEHCIGIKPCSEE
 GTRKLVRAAIEYAIANDRDSVTLVHKGNIMKFTGAFKDWGYQLAREEFGG
 ELIDGGPWLKVNPNNTGKEIVIKDVIADAFLOQIILLRPAEYDVIA CMNLNG
 DYISDALAAQVGGIGIAPGANIGDECALFEATHGTAPKYAGQDKVNP GSII
 LSAEMMLRHMGTAEADLIVKMEGAINAKTVTYDFERLMDGAKLLKCSF
 GDIIENM

Umbelopsis isabellina MI2 nucleotide sequence

(SEQ ID NO: 5)

ATGAGCACCACCACCTGATTGTGGCCATTAAGGGTACCGATAATGTTGAA
 CGTGATGGCGCAATTGTTCTGCTGGATGCCGGCGCCAGTATGGAACCCCTG
 CGTCCGCGCATTCAGAAAACTGGCCATTAGCAGTGGCATTGAAGATCTG
 ATTCTGGAAGATGCAATGGTGACAATCTGACCACCATTGATCAGGTTCTGT
 AAACAGCAGACCGTTTTTGTTAATCTGGAAGATCAGATTAAAGCTGCCGGCA
 GTTCCGCGCCCATACCCTGCCGTATTTTGGTAATCTGTATCAGCTGCTGCCG
 GATATGCTGGCCGGTTGGCGCAAACCTGTTTGTATGAATATGGTCCGGTTGTT
 AAAGTGAATCTGCTGGGCAATGAAATTATTGGTACCAATGATCCGGCCGTG
 GCCGAACGTGGGTTAAAGAAAGCGAATATTTTACCAAAAAGATCTACGGT
 GGCGTGCAGGAAGTTAAAGTTTTGGCGGCCAGGGTCTGTTTACCACCGAT
 AGCGATGATATGGATTGGAACTGGCACATAAACTGCTGATGCCGGCATT
 TCACCGCGCGCCATTAAAGTTTATCAGCATGAAATGAGCGTGATTGGTCTG
 CAGACCATTAAAGTTTTTGAACAGTATAGCCCGGATGAAGAAGTGGAAT

35

-continued

CTGCATTGGACCACCAATCTGACCTTTGAAACCATTGGCAAAGTTGGTTTT
 GGCTATGATTTTCATCTGCTGGATGATCGTTATGGCGAAATCATCCGTTT
 ATTGAAGCAATGGGTTATTGCATGAAACAGAGCTTTGCCCGTGGCACCCAG
 AGCAAACGATTAAGTATCTGCCGATTGAAGCCAATCGTCGCTATGATCGC
 AGTCTGAATCTGATGCATAGCATTTGTTGATGAAGTTATTACCCAGCGTAA
 AGCCATCCGCATGCAAGCGAAGATAATAAGGATCTGCTGGATTTTATGCTG
 ACCGCACGTGATGAAAATAATCTGGGCCTGAGCGATGAAAATATTGCGGAT
 CAGGTTATTACCTTTCTGATTGCCGGCCATGAAACCACCAGCAATACCCTG
 GCATGGACCCGTGATGAACGAGTCGCCATCCGGAATGAACAGAAAATT
 CTGCAGGAAGTGGTGAATCTGGGCATTACCACCGATTCACTGCCGACCAGC
 GAACGAGAGCAGTAGCATGAAATATACCTATCAGGTGCTGAAAGAAACCCTG
 CGCATGTATAGTCCGCTGCGTGACATGGCAAAATATTGCAAAAAAGATATT
 GTGGTGCCGGGCGGCTATCAGATTAAGGCAGGCGATCGTGTGGCCGTTT
 CTGAATAGCCTGCATTATAATGAAAAAGTTTACCCGAATCCGACCCAGTAT
 GATCCGAGTCGCTGGACCCCGAAGAAGAACAGAAACGAGTCGCTTTGCC
 TGGCTGCCGTTTATGACACCGGCCCGCGTAGCTGCATTGGTATGGCACTGGCC
 CTGCAGGAAGCAAAAACCATTTCTGGCAATGATTCTGCTGAAATTTGCTTT
 GTTTATGATGGTCCGCCGATTGGTTATGATCCGAAAAGCCCGACCATTCGT
 CCGCTGAATCTGATGATGAAAATTTCTGCCGCGTGATAAACTGCCGAGCCCG
 ACCGCCGATAATAAGCTGACCCCGGTTGGCAGTCCGAATATTGGTAAAGCC
 CAGATGGCAATGCCGACCGCAGCCACCAATGTTGGTACCGCCGAATGCCG
 CCGGTTTTCTTTCTGTATGGTACCCAGACCGGTACCGCCAGGATTATGCA
 AGTCAGCTGGCAAGCCAGGCCAAAAGTTTGGTTTTAAAGATATTACCCTG
 TGTGAAATGGATAAATGGGAAGTGTGCGAGCGCAAAATATACCGGTGCC
 AAAGAAGAAAAAGATACCCGTGCCCTGGTTGTTATTGTACCGCAACCTAT
 AATGGCAGCCCGCCGATAGTCAGAAAAATTTGATAAATTCATTACCGAC
 ACCAGCAAAGATAATGATCTGCCGCTGAAAGGCCCTGCTGTATACCGTGT
 GGCTTGGCAATCGTAATTGGCGCACCTATCAGCAGTTTCCGATTAAGTGC
 GATGCCCGTCTGGATGAACGGTGGTGAACGCTTTTATGATCTGGGCAGC
 GGCAATGCCGATAAAGATATGGATGGTGAATTCATGATTGGTGCGCCCAT
 TTTTGGACCCATACCCCTGACCTATTATGGCATTGCAACCCATGAAGGTAGT
 GCAAGTCTGGTGGCGGCCAGAAAACCCAGAGATCCGACCGCGGTCTG
 GAAATTCGTTTTATTCCGCCGAGCGAAAGCGAAGCCTGGGATAAAGCCGAT
 AAAAAATGTTAATGGTGACTATAATGCGGAATTCAGGTGAATCGCGAACTG
 CAGAATATTGAACGTAGTAAACGTAGCACCCGCCATATTGAAATTGATATT
 AGTAAACTGCAGAGCCCGATACCGATAAACATCGTTATCTGACCGGTGAC
 CATCTGGAAGTGATCCGGAATGCGGATAATGTGGTGAAAGCATTGGCC
 CTGGGCTTTGGTCTGATTCTGGATAGTGTGTTTGAATACCAGCGTGGAT

36

-continued

AAAAGTACCGTGAGTAGTCGTAGCCTGGCAGCCAGTATTACCGGCCCGTGT
 ACCGTTTCGTAATGCCCTGAAATATTATGCCGATATATATAGTGCCCGGAGC
 CGTTATCTGCTGGCCTTTTTTCGCAGCACGCTGCAGGATACCCATCCGGAT
 GTTGCCGCAAGTTTTAGCGATGTGATTATTCGGGTGAAAAATGGCCAGAAA
 GCCTATAATGAATTCATTTCAGAAATACCGTAACCTGCTGGATCTGCAGCGC
 GGCTTTCCGCTGAAAGAACTGGATCTGAAAGAATTTCTGTGCGCAGTTAGC
 GTTATGCAGCGCGCGCGTTATAGCATTGCCAGCAGCCCGCTGAAAGCAGAA
 GATACCGCATATCTGGCAGTTGGTGTGGTGGATGATGTGTTTGCAAATCGT
 CATTATTATGGCCTGGCCAGTGGCTATCTGGCCCGCAGCCAGCCGCCGACA
 CCGATTCTGTCACGTATTAAGAGTAGCAAAAGTACCTTTGGCCTGCCGGAA
 AATCCGGAACCCCGATTATTATGATTGGTGCCGGTACCGGTATTAGTCCG
 TTTATGGGCTTTATGCAGGAACGCGAAATGCTGAAAGCAAGCAGAAGCC
 CATCTGTTTTTCGGCTGCCGCCATCCGGATGAAGATTTTATCTATCTAGC
 GTTTTTGAAGGCTATGAAAAAGCGCGTTATTACCAAATGTATCCGGCC
 TTTAGCCGCTCTGGGTGAAGATAATCCGCGCAAATATGTTGAGCATCAGCTG
 ATGGCAAATGCCGCGCAAATTTGGACCTGCTGACCAATGGCGCAAATGTG
 TATGTTTGGCGCAGTGGCCCGATGAGCCCGCATGTGCGTCGTAGCTTTGAA
 CTGATGGCAAAAAGTTTTTGAGGCGCCAAAACCGATGATGAAGCATGCGAA
 AAATGCTGGAATTCATGAGCGCAGGTCGTTATAATGAAGATGTGTGGGGT
 TAA
 35 *Umbelopsis isabellina* MI2 amino acid sequence (SEQ ID NO: 6)
 MSTTTLIVAIKGTNDNERDGAIVRLDAGASMETLRPRIAEKLAISSGIEDL
 ILEDANGDNLTTIDQVRKQQTVPVFNLEDQIKLPAVPAHTLPYPGNLYQLLP
 DMLAGWRKLFDEYGPVVKVNLGNEIIGTNDPAVAELVWKESEYFTKKIYG
 40 GLQEVKSFQGQLFTTDSDDMDWLAKHLMLPAFSPRAIKVQHEMSVIGL
 QTIKVFQYSPDEEVEILHWTNLTFTETIGKVFGYDFHLDDRYGENHPF
 IEAMGYCMKQSFARGTQSKLIKYLPIEANRRYDRSLNLMHSIVDEVITQRK
 45 SHPHASEDNKDLLDFMLTARDENNGLSDENIRQVITFLIAGHETTSNTL
 AWTLYELSRHPEIEQKILQEVNLTGTTDLSLPTSEQSSSMKYTYQVLKETL
 RMYSLRALAKYCKKDIVVPGYQIKAGDRVAVQLNSLHYNEKVYPNPTQY
 50 DPSRWTPEEEQKRSRFAWLPFSTGPRSCIGMALALQEAKTILAMILLKFRF
 VYDGPPIGYDPKSPTIRPLNLMKILPRDKLPSPTADNKLTPVGSFNIGKA
 QMAMPTAATNVGTAEPLPVFFLYGTQGTAGDQYASQLASQAKSFGFKDITL
 55 CEMDKWEVLQSGKYTGAKKEKDRALVVICTATYNGSPPDSEAFNDFKFTD
 TSKDNDLPLKGLLYTVFGLGNRNWRITYQFPFIKCDARLDELGGERFYDLGS
 GNADKMDGEFHDWCAHFWTHLTYYGIATHEGSASLVPGQKTTEDPTSGL
 60 EIRFIPPSESEAWDKADKNVNGDYNAEIQVNRLEQNIERSKRSTRHIEIDI
 SKLQSPDTHRYLTGDHLEVYPENADNVVESIALGFLILDSVFEITSVD
 KSTVSSRSLAASITGPCTVRNALKYYADIYSAPSRYLALFAARLQDTHPD
 65 VAASFSDVPIPGENGQKAYNEFIQKYRNLLDLQRGFPLKELDLKEFLCAVS

37

-continued

VMQPRRYSIASSPLKAEDTAYLAVGVVDDVFNRRHYGLASGYLARSQPPT

PIRARIKSSKSTFGLPENPETPIIMIGAGTGISPFMGFMQEREMLSKSAEA

HLFFGCRHPDEDFIYRSVFEGYEKSGVITKLYPAFSRLGEDNPRKYVQHQL

MANAGKIWTLLTNGANVYVCGSGPMSRDVRRSFELMAKSPFGAKTDEACE

KLEFMSAGRYNEDVWG

ICD-F Primer

(SEQ ID No: 7)

GGAATTCATATGAAAGTAAAGTAGTTGT

ICD-R Primer

(SEQ ID No: 8)

CCGCTCGAGTTACATGTTTTCGATGATCGCGT

38

TABLE 2-continued

Sequence Information	
Seq ID No.	Sequence Description
4	icd protein from <i>E. coli</i> K-12 substr. MG1655
5	<i>Umbelopsis isabellina</i> MI2 BM3 homolog nucleotide sequence codon-optimization for <i>E. coli</i> expression
6	<i>Umbelopsis isabellina</i> MI2 BM3 homolog amino acid sequence

TABLE 1

Examples of fatty acids			
Common Name	Systematic Name	Structural Formula	Lipid Numbers
Valeric acid	Pentanoic acid	CH ₃ (CH ₂) ₃ COOH	C5:0
Caproic acid	Hexanoic acid	CH ₃ (CH ₂) ₄ COOH	C6:0
Enanthic acid	Heptanoic acid	CH ₃ (CH ₂) ₅ COOH	C7:0
Caprylic acid	Octanoic acid	CH ₃ (CH ₂) ₆ COOH	C8:0
Pelargonic acid	Nonanoic acid	CH ₃ (CH ₂) ₇ COOH	C9:0
Capric acid	Decanoic acid	CH ₃ (CH ₂) ₈ COOH	C10:0
Undecylic acid	Undecanoic acid	CH ₃ (CH ₂) ₉ COOH	C11:0
Lauric acid	Dodecanoic acid	CH ₃ (CH ₂) ₁₀ COOH	C12:0
Tridecylic acid	Tridecanoic acid	CH ₃ (CH ₂) ₁₁ COOH	C13:0
Myristic acid	Tetradecanoic acid	CH ₃ (CH ₂) ₁₂ COOH	C14:0
Pentadecylic acid	Pentadecanoic acid	CH ₃ (CH ₂) ₁₃ COOH	C15:0
Palmitic acid	Hexadecanoic acid	CH ₃ (CH ₂) ₁₄ COOH	C16:0
Margaric acid	Heptadecanoic acid	CH ₃ (CH ₂) ₁₅ COOH	C17:0
Stearic acid	Octadecanoic acid	CH ₃ (CH ₂) ₁₆ COOH	C18:0
Nonadecylic acid	Nonadecanoic acid	CH ₃ (CH ₂) ₁₇ COOH	C19:0
Arachidic acid	Eicosanoic acid	CH ₃ (CH ₂) ₁₈ COOH	C20:0
Heneicosylic acid	Heneicosanoic acid	CH ₃ (CH ₂) ₁₉ COOH	C21:0
Behenic acid	Docosanoic acid	CH ₃ (CH ₂) ₂₀ COOH	C22:0
Tricosylic acid	Tricosanoic acid	CH ₃ (CH ₂) ₂₀ COOH	C23:0
Lignoceric acid	Tetracosanoic acid	CH ₃ (CH ₂) ₂₂ COOH	C24:0
Pentacosylic acid	Pentacosanoic acid	CH ₃ (CH ₂) ₂₃ COOH	C25:0
Cerotic acid	Hexacosanoic acid	CH ₃ (CH ₂) ₂₄ COOH	C26:0
Heptacosylic acid	Heptacosanoic acid	CH ₃ (CH ₂) ₂₅ COOH	C27:0
Montanic acid	Octacosanoic acid	CH ₃ (CH ₂) ₂₆ COOH	C28:0
Nonacosylic acid	Nonacosanoic acid	CH ₃ (CH ₂) ₂₇ COOH	C29:0
Melissic acid	triacontanoic acid	CH ₃ (CH ₂) ₂₈ COOH	C30:0
Henatriacontylic acid	Henatriacontanoic acid	CH ₃ (CH ₂) ₂₉ COOH	C31:0
Lacceroic acid	Dotriacontanoic acid	CH ₃ (CH ₂) ₃₀ COOH	C32:0
Psyllic acid	Tritriacontanoic acid	CH ₃ (CH ₂) ₃₁ COOH	C33:0
Geddic acid	Tetatriacontanoic acid	CH ₃ (CH ₂) ₃₂ COOH	C34:0
Ceroplastic acid	Pentatriacontanoic acid	CH ₃ (CH ₂) ₃₃ COOH	C35:0
Hexatriacontylic acid	Hexatriacontanoic acid	CH ₃ (CH ₂) ₃₄ COOH	C36:0
Heptatriacontanoic acid	Heptatriacontanoic acid	CH ₃ (CH ₂) ₃₅ COOH	C37:0
Octatriacontanoic acid	Octatriacontanoic acid	CH ₃ (CH ₂) ₃₆ COOH	C38:0
Nonatriacontanoic acid	Nonatriacontanoic acid	CH ₃ (CH ₂) ₃₇ COOH	C39:0
Tetracontanoic acid	Tetracontanoic acid	CH ₃ (CH ₂) ₃₈ COOH	C40:0

TABLE 2

Sequence Information	
Seq ID No.	Sequence Description
1	<i>M. thermophile</i> CYP505A30 nucleotide sequence with codon-optimization for <i>E. coli</i>
2	icd gene from <i>E. coli</i> K-12 substr. MG1655
3	<i>M. thermophile</i> CYP505A30 amino acid sequence

TABLE 2-continued

Sequence Information	
Seq ID No.	Sequence Description
7	ICD-F Primer 5'GGAATTCATATGAAAGTAAAGTAGTTGT3'
8	ICD-R Primer 5'CCGCTCGAGTTACATGTTTTCGATGATCGCGT3'

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 8

<210> SEQ ID NO 1

<211> LENGTH: 3243

<212> TYPE: DNA

<213> ORGANISM: Myceliophthora thermophila

<400> SEQUENCE: 1

```

atggcggata agaccaccca aaccgtgccg attccgggtc cgcggggcct gccgctggtt    60
ggtaatgcgc tggcggttga tagcgaaactg ccgctgcgta ccttcaggga atttgcggag    120
gaatacggcg agatctatcg tctgaccctg ccgaccggta ccacctgggt ggtagcagc    180
caagcgtctg ttcacgaact gtgcgacgat aagcgtttca agaagccggt tgcgcggcg    240
ctggcggaag tgcgtaacgg cgtaaacgac ggtctgttta ccgcgcgtga agaggagccg    300
aactggggca tcgcgcaccg tattctgatg ccggcggttg gtccggcgag cattcagggc    360
atgtttaccg aaatgcacga gatcgcgagc caactggcgc tgaaatgggc gcgtcacggt    420
ccggacaccc cgattttctg taccgacgat ttaccctgc tgacctgga taccctggcg    480
ctgtgcacca tgaacttcgg ttttaacagc tactatcacg acgaactgca cccgttcac    540
aacgcgatgg gcaactttct gaccgagagc ggtgcgcgtg cgatgcgtcc ggcgatcacc    600
agcattttcc accaggcggc gaaccgtaag tactgggaag atattgaggt tctgcgtaaa    660
accgcgcaag gtgtgctgga caccctcgtt aagcaccgca ccaaccgtaa agatctgctg    720
agcgcgatgc tggacggcgt ggatgcgaaa accggtcaga aactgagcga cagcagcatc    780
attgataacc tgatcacctt tctgattgcg ggccacgaaa ccaccagcgg tctgctgagc    840
ttcgcgtttt acctgctgat taagcaccag gacgcgtatc gtaaagcgca agaagagggtg    900
gatcgtgtta tcggcaaggg cccgattaaa gttgaacaca tcaagaaact gccgtacatc    960
gcggcggtgc tgcgtgaaac cctgcgtctg tgcccagaca ttccgatcat taaccgtgcg    1020
gcgaagcagg acgaagtatt cggtggaag tacgcgggtg cgaaagatca gcgtctggcg    1080
ctgctgctgg cgcaaaagcca cctggaccgg gcggtttatg gcgaaaccgc gaagcaatc    1140
attccggagc gtatgctgga cgaaaacttt gagcgtctga accgtgagta tccgattgct    1200
tggaaacctg tcggtaccgg catgcgtgctg tgcacgggtc gtccgtttgc gtggcaggaa    1260
ggcgtgctgg ttatggcgat gctgctgcaa aacttcgact ttgttctgca cgatccgtac    1320
tatgagctgc actacaagca gacctgacc accaagccga aagacttcta tatgctgctg    1380
atcctgcgtg atggcctgac cgcgaccgaa ctggagcacc gtctggcggg taacgcggcg    1440
agcgtggcgc gtacgggtgg cgggtggcgt ggcccgagca aaccgaccgc gcagaaaacc    1500
agcccgccgg aagcgaaacc gatgagcatc ttctacggca gcaacaccgg tacctgcgag    1560
agcctggcgc aacgtctggc gaccgatgct gcgagccacg gttatgctgc ggcgggcgtg    1620
gaaccgctgg acaccgcgac cgagaagctg ccgaccgata gtccgggtgt tatcattacc    1680
gcgagcttcg agggctcagc gccggacaac gcggcgaggt tttgcggctg gctgaaaaac    1740
ctggaagggt atgagctgaa aaactgagc tacgcgggtt tcggttgctg ccaccacgac    1800
tggaaccaga cctttcaccg tattccgaag ctggttcacc aaaccatgaa agcgcacggt    1860
gcgagcccca tctgcgacga agcctgacc gatgtggcgg agggtaacat gttcaccgat    1920
tttgaacaat gggaggacga tgtgttctgg ccggcggttc gtgcgcgtta tggcgcgcg    1980
ggtgcggttg cggaaaccga ggacgcgccg ggtagcgatg gtctgaacat ccactttagc    2040
agcccgcgta gcagaccct gcgtcaggac gtgcgtgaag cgaccgtggt tggatgaagc    2100

```

-continued

ctgctgaccg cgccggatgc gccgccgaag aaacacattg aagttcaact gccggacggc	2160
gcgacctaca aagtgggtga ttatctggcg gtgctgccgg ttaacagcaa ggagagcatt	2220
ggtcgtgtta tgcgtaaatt ccagctgagc tgggacagcc acgtgaccat cgcgagcgat	2280
cgttggaccg cgctgccgac cggtaacccg gtgccggcgt acgacgttct gggtagctat	2340
gtggagctga gccaacccgg gaccaaactg ggtatcctgc gtctggcgga tgcggcgga	2400
gatgaggcga ccaaggcgga actgcaaaaa ctggcgggtg atctgtacac cagcgagatt	2460
agcctgaaac gtgcgagcgt tctggacctg ctggatcgtt tcccgagcat cagcctgccg	2520
ttcggtaact ttctgagcct gctgccgccg attcgtccgc gtcaatacag catcagcagc	2580
agcccgtga acgacccgag ccgtgcgacc ctgacctata gcctgctgga tagcccgagc	2640
ctggcgaacc cgagccgtcg ttctgtgggc gttgcgacca gctacctgag cagcctggtt	2700
cgtggtgaca agctgctggt gagcgttcgt ccgacccaca ccgcgtttcg tctgccggac	2760
gaagataaaa tgggtgaaac cgcgatcatt tgcgtgggtg cgggtagcgg tctggcgccg	2820
ttccgtggtt ttatccagga acgtgcggcg ctgctggcga aaggtaccca actggcgcg	2880
gcgctgctgt tctacggttg ccgtagcccg gagaaggacg atctgtatcg tgacgaattc	2940
gataaatggc aagagagcgg tgcggtggat gttcgtcgtg cgtttagcgg tgttgatagc	3000
gacgataccg aggcgcgtgg ttgccgtcac gttcaggacc gtctgtggca cgatcgtgaa	3060
gaggtgaagg cgctgtggga ccgtggcgcg cgtgtgtacg tttgcggtag ccgtcaagtg	3120
ggcgaagggt ttaaaaccgc gatgggccgt atcgtgctgg gtgaagagga cgcggaggat	3180
gcgatcagca agtgggtatga aaccgtgcgt aatgaccgtt atgcgaccga tgtgttcgac	3240
taa	3243

<210> SEQ ID NO 2

<211> LENGTH: 1251

<212> TYPE: DNA

<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 2

atggaaagta aagtagttgt tccggcaciaa ggcaagaaga tcacctgca aaacggcaaa	60
ctcaacgttc ctgaaaatcc gattatccct tacattgaag gtgatggaat cggtgtagat	120
gtaaccccg ccatgctgaa agtggtcgac gctgcagtcg agaaagccta taaaggcgag	180
cgtaaaaatc cctggatgga aatttacacc ggtgaaaaat ccacacaggt ttatggtcag	240
gacgtctggc tgctgctga aactcttgat ctgattcgtg aatatcgcgt tgccattaaa	300
ggtccgctga ccactccggt tggtgccggt attcgtctc tgaacgttgc cctgcgccag	360
gaactggatc tctacatctg cctgcgtccg gtacgttact atcagggcac tccaagcccg	420
gttaaacacc ctgaactgac cgatatgggt atcttccgtg aaaactcggg agacatttat	480
gctgggtatc aatggaaagc agactctgcc gacgccgaga aagtgattaa attcctgcgt	540
gaagagatgg ggggtaagaa aattcgcttc ccggaacatt gtggtatcgg tattaagccg	600
tgttcggaag aaggcaccac acgtctggtt cgtgcagcga tcgaatacgc aattgctaac	660
gatcgtgact ctgtgactct ggtgcacaaa ggcaacatca tgaagttcac cgaaggagcg	720
tttaagact ggggctacca gctggcgcggt gaagagtttg gcggtgaact gatcgacggt	780
ggcccgctggc tgaaggttaa aaacccgaac actggcaaaag agatcgtcat taaagacgtg	840
attgctgatg cattcctgca acagatcctg ctgcgtccgg ctgaatatga tgttatcgcc	900

-continued

```

tgtatgaacc tgaacggtga ctacatttct gacgccctgg cagcgcaggt tggcggtatc   960
ggatcgcccc ctggtgcaaa catcggtgac gaatcgcccc tggttgaagc caccacggt   1020
actgcgccga aatatgccgg tcaggacaaa gtaaactcctg gctctattat tctctcggct   1080
gagatgatgc tgcgccacat gggttggacc gaagcggctg acttaattgt taaaggtatg   1140
gaaggcgcaa tcaacgcgaa aaccgtaacc tatgacttcg agcgtctgat ggatggcgct   1200
aaactgctga aatgttcaga gtttgggtgac gcgatcatcg aaaacatgta a           1251

```

```

<210> SEQ ID NO 3
<211> LENGTH: 1080
<212> TYPE: PRT
<213> ORGANISM: Myceliophthora thermophila

```

```

<400> SEQUENCE: 3

```

```

Met Ala Asp Lys Thr Thr Glu Thr Val Pro Ile Pro Gly Pro Gly
1      5      10      15
Leu Pro Leu Val Gly Asn Ala Leu Ala Phe Asp Ser Glu Leu Pro Leu
20     25     30
Arg Thr Phe Gln Glu Phe Ala Glu Glu Tyr Gly Glu Ile Tyr Arg Leu
35     40     45
Thr Leu Pro Thr Gly Thr Thr Leu Val Val Ser Ser Gln Ala Leu Val
50     55     60
His Glu Leu Cys Asp Asp Lys Arg Phe Lys Lys Pro Val Ala Ala Ala
65     70     75     80
Leu Ala Glu Val Arg Asn Gly Val Asn Asp Gly Leu Phe Thr Ala Arg
85     90     95
Glu Glu Glu Pro Asn Trp Gly Ile Ala His Arg Ile Leu Met Pro Ala
100    105    110
Phe Gly Pro Ala Ser Ile Gln Gly Met Phe Thr Glu Met His Glu Ile
115    120    125
Ala Ser Gln Leu Ala Leu Lys Trp Ala Arg His Gly Pro Asp Thr Pro
130    135    140
Ile Phe Val Thr Asp Asp Phe Thr Arg Leu Thr Leu Asp Thr Leu Ala
145    150    155    160
Leu Cys Thr Met Asn Phe Arg Phe Asn Ser Tyr Tyr His Asp Glu Leu
165    170    175
His Pro Phe Ile Asn Ala Met Gly Asn Phe Leu Thr Glu Ser Gly Ala
180    185    190
Arg Ala Met Arg Pro Ala Ile Thr Ser Ile Phe His Gln Ala Ala Asn
195    200    205
Arg Lys Tyr Trp Glu Asp Ile Glu Val Leu Arg Lys Thr Ala Gln Gly
210    215    220
Val Leu Asp Thr Arg Arg Lys His Pro Thr Asn Arg Lys Asp Leu Leu
225    230    235    240
Ser Ala Met Leu Asp Gly Val Asp Ala Lys Thr Gly Gln Lys Leu Ser
245    250    255
Asp Ser Ser Ile Ile Asp Asn Leu Ile Thr Phe Leu Ile Ala Gly His
260    265    270
Glu Thr Thr Ser Gly Leu Leu Ser Phe Ala Phe Tyr Leu Leu Ile Lys
275    280    285
His Gln Asp Ala Tyr Arg Lys Ala Gln Glu Glu Val Asp Arg Val Ile
290    295    300
Gly Lys Gly Pro Ile Lys Val Glu His Ile Lys Lys Leu Pro Tyr Ile
305    310    315    320

```

-continued

Ala	Ala	Val	Leu	Arg	Glu	Thr	Leu	Arg	Leu	Cys	Pro	Thr	Ile	Pro	Ile	
				325						330					335	
Ile	Asn	Arg	Ala	Ala	Lys	Gln	Asp	Glu	Val	Ile	Gly	Gly	Lys	Tyr	Ala	
			340					345						350		
Val	Ala	Lys	Asp	Gln	Arg	Leu	Ala	Leu	Leu	Leu	Ala	Gln	Ser	His	Leu	
		355						360					365			
Asp	Pro	Ala	Val	Tyr	Gly	Glu	Thr	Ala	Lys	Gln	Phe	Ile	Pro	Glu	Arg	
	370						375					380				
Met	Leu	Asp	Glu	Asn	Phe	Glu	Arg	Leu	Asn	Arg	Glu	Tyr	Pro	Asp	Cys	
385					390					395					400	
Trp	Lys	Pro	Phe	Gly	Thr	Gly	Met	Arg	Ala	Cys	Ile	Gly	Arg	Pro	Phe	
			405						410						415	
Ala	Trp	Gln	Glu	Ala	Val	Leu	Val	Met	Ala	Met	Leu	Leu	Gln	Asn	Phe	
			420					425						430		
Asp	Phe	Val	Leu	His	Asp	Pro	Tyr	Tyr	Glu	Leu	His	Tyr	Lys	Gln	Thr	
		435					440						445			
Leu	Thr	Thr	Lys	Pro	Lys	Asp	Phe	Tyr	Met	Arg	Ala	Ile	Leu	Arg	Asp	
	450					455					460					
Gly	Leu	Thr	Ala	Thr	Glu	Leu	Glu	His	Arg	Leu	Ala	Gly	Asn	Ala	Ala	
465					470					475					480	
Ser	Val	Ala	Arg	Ser	Gly	Gly	Gly	Gly	Gly	Gly	Pro	Ser	Lys	Pro	Thr	
			485					490						495		
Ala	Gln	Lys	Thr	Ser	Pro	Ala	Glu	Ala	Lys	Pro	Met	Ser	Ile	Phe	Tyr	
			500					505						510		
Gly	Ser	Asn	Thr	Gly	Thr	Cys	Glu	Ser	Leu	Ala	Gln	Arg	Leu	Ala	Thr	
		515					520					525				
Asp	Ala	Ala	Ser	His	Gly	Tyr	Ala	Ala	Ala	Ala	Val	Glu	Pro	Leu	Asp	
	530					535					540					
Thr	Ala	Thr	Glu	Lys	Leu	Pro	Thr	Asp	Arg	Pro	Val	Val	Ile	Ile	Thr	
545					550					555					560	
Ala	Ser	Phe	Glu	Gly	Gln	Pro	Pro	Asp	Asn	Ala	Ala	Lys	Phe	Cys	Gly	
			565					570						575		
Trp	Leu	Lys	Asn	Leu	Glu	Gly	Asp	Glu	Leu	Lys	Asn	Val	Ser	Tyr	Ala	
			580					585					590			
Val	Phe	Gly	Cys	Gly	His	His	Asp	Trp	Ser	Gln	Thr	Phe	His	Arg	Ile	
	595						600					605				
Pro	Lys	Leu	Val	His	Gln	Thr	Met	Lys	Ala	His	Gly	Ala	Ser	Pro	Ile	
	610					615					620					
Cys	Asp	Glu	Gly	Leu	Thr	Asp	Val	Ala	Glu	Gly	Asn	Met	Phe	Thr	Asp	
625					630					635					640	
Phe	Glu	Gln	Trp	Glu	Asp	Asp	Val	Phe	Trp	Pro	Ala	Val	Arg	Ala	Arg	
			645						650					655		
Tyr	Gly	Ala	Ala	Gly	Ala	Val	Ala	Glu	Thr	Glu	Asp	Ala	Pro	Gly	Ser	
		660						665					670			
Asp	Gly	Leu	Asn	Ile	His	Phe	Ser	Ser	Pro	Arg	Ser	Ser	Thr	Leu	Arg	
		675						680					685			
Gln	Asp	Val	Arg	Glu	Ala	Thr	Val	Val	Gly	Glu	Ala	Leu	Leu	Thr	Ala	
	690					695					700					
Pro	Asp	Ala	Pro	Pro	Lys	Lys	His	Ile	Glu	Val	Gln	Leu	Pro	Asp	Gly	
705					710					715					720	
Ala	Thr	Tyr	Lys	Val	Gly	Asp	Tyr	Leu	Ala	Val	Leu	Pro	Val	Asn	Ser	
			725						730						735	

-continued

Lys Glu Ser Ile Gly Arg Val Met Arg Lys Phe Gln Leu Ser Trp Asp
 740 745 750
 Ser His Val Thr Ile Ala Ser Asp Arg Trp Thr Ala Leu Pro Thr Gly
 755 760 765
 Thr Pro Val Pro Ala Tyr Asp Val Leu Gly Ser Tyr Val Glu Leu Ser
 770 775 780
 Gln Pro Ala Thr Lys Arg Gly Ile Leu Arg Leu Ala Asp Ala Ala Glu
 785 790 795 800
 Asp Glu Ala Thr Lys Ala Glu Leu Gln Lys Leu Ala Gly Asp Leu Tyr
 805 810 815
 Thr Ser Glu Ile Ser Leu Lys Arg Ala Ser Val Leu Asp Leu Leu Asp
 820 825 830
 Arg Phe Pro Ser Ile Ser Leu Pro Phe Gly Thr Phe Leu Ser Leu Leu
 835 840 845
 Pro Pro Ile Arg Pro Arg Gln Tyr Ser Ile Ser Ser Ser Pro Leu Asn
 850 855 860
 Asp Pro Ser Arg Ala Thr Leu Thr Tyr Ser Leu Leu Asp Ser Pro Ser
 865 870 875 880
 Leu Ala Asn Pro Ser Arg Arg Phe Val Gly Val Ala Thr Ser Tyr Leu
 885 890 895
 Ser Ser Leu Val Arg Gly Asp Lys Leu Leu Val Ser Val Arg Pro Thr
 900 905 910
 His Thr Ala Phe Arg Leu Pro Asp Glu Asp Lys Met Gly Glu Thr Ala
 915 920 925
 Ile Ile Cys Val Gly Ala Gly Ser Gly Leu Ala Pro Phe Arg Gly Phe
 930 935 940
 Ile Gln Glu Arg Ala Ala Leu Leu Ala Lys Gly Thr Gln Leu Ala Ala
 945 950 955 960
 Ala Leu Leu Phe Tyr Gly Cys Arg Ser Pro Glu Lys Asp Asp Leu Tyr
 965 970 975
 Arg Asp Glu Phe Asp Lys Trp Gln Glu Ser Gly Ala Val Asp Val Arg
 980 985 990
 Arg Ala Phe Ser Arg Val Asp Ser Asp Asp Thr Glu Ala Arg Gly Cys
 995 1000 1005
 Arg His Val Gln Asp Arg Leu Trp His Asp Arg Glu Glu Val Lys
 1010 1015 1020
 Ala Leu Trp Asp Arg Gly Ala Arg Val Tyr Val Cys Gly Ser Arg
 1025 1030 1035
 Gln Val Gly Glu Gly Val Lys Thr Ala Met Gly Arg Ile Val Leu
 1040 1045 1050
 Gly Glu Glu Asp Ala Glu Asp Ala Ile Ser Lys Trp Tyr Glu Thr
 1055 1060 1065
 Val Arg Asn Asp Arg Tyr Ala Thr Asp Val Phe Asp
 1070 1075 1080

<210> SEQ ID NO 4

<211> LENGTH: 416

<212> TYPE: PRT

<213> ORGANISM: Escherichia coli

<400> SEQUENCE: 4

Met Glu Ser Lys Val Val Val Pro Ala Gln Gly Lys Lys Ile Thr Leu
 1 5 10 15
 Gln Asn Gly Lys Leu Asn Val Pro Glu Asn Pro Ile Ile Pro Tyr Ile
 20 25 30

-continued

Glu Gly Asp Gly Ile Gly Val Asp Val Thr Pro Ala Met Leu Lys Val
 35 40 45
 Val Asp Ala Ala Val Glu Lys Ala Tyr Lys Gly Glu Arg Lys Ile Ser
 50 55 60
 Trp Met Glu Ile Tyr Thr Gly Glu Lys Ser Thr Gln Val Tyr Gly Gln
 65 70 75 80
 Asp Val Trp Leu Pro Ala Glu Thr Leu Asp Leu Ile Arg Glu Tyr Arg
 85 90 95
 Val Ala Ile Lys Gly Pro Leu Thr Thr Pro Val Gly Gly Gly Ile Arg
 100 105 110
 Ser Leu Asn Val Ala Leu Arg Gln Glu Leu Asp Leu Tyr Ile Cys Leu
 115 120 125
 Arg Pro Val Arg Tyr Tyr Gln Gly Thr Pro Ser Pro Val Lys His Pro
 130 135 140
 Glu Leu Thr Asp Met Val Ile Phe Arg Glu Asn Ser Glu Asp Ile Tyr
 145 150 155 160
 Ala Gly Ile Glu Trp Lys Ala Asp Ser Ala Asp Ala Glu Lys Val Ile
 165 170 175
 Lys Phe Leu Arg Glu Glu Met Gly Val Lys Lys Ile Arg Phe Pro Glu
 180 185 190
 His Cys Gly Ile Gly Ile Lys Pro Cys Ser Glu Glu Gly Thr Lys Arg
 195 200 205
 Leu Val Arg Ala Ala Ile Glu Tyr Ala Ile Ala Asn Asp Arg Asp Ser
 210 215 220
 Val Thr Leu Val His Lys Gly Asn Ile Met Lys Phe Thr Glu Gly Ala
 225 230 235 240
 Phe Lys Asp Trp Gly Tyr Gln Leu Ala Arg Glu Glu Phe Gly Gly Glu
 245 250 255
 Leu Ile Asp Gly Gly Pro Trp Leu Lys Val Lys Asn Pro Asn Thr Gly
 260 265 270
 Lys Glu Ile Val Ile Lys Asp Val Ile Ala Asp Ala Phe Leu Gln Gln
 275 280 285
 Ile Leu Leu Arg Pro Ala Glu Tyr Asp Val Ile Ala Cys Met Asn Leu
 290 295 300
 Asn Gly Asp Tyr Ile Ser Asp Ala Leu Ala Ala Gln Val Gly Gly Ile
 305 310 315 320
 Gly Ile Ala Pro Gly Ala Asn Ile Gly Asp Glu Cys Ala Leu Phe Glu
 325 330 335
 Ala Thr His Gly Thr Ala Pro Lys Tyr Ala Gly Gln Asp Lys Val Asn
 340 345 350
 Pro Gly Ser Ile Ile Leu Ser Ala Glu Met Met Leu Arg His Met Gly
 355 360 365
 Trp Thr Glu Ala Ala Asp Leu Ile Val Lys Gly Met Glu Gly Ala Ile
 370 375 380
 Asn Ala Lys Thr Val Thr Tyr Asp Phe Glu Arg Leu Met Asp Gly Ala
 385 390 395 400
 Lys Leu Leu Lys Cys Ser Glu Phe Gly Asp Ala Ile Ile Glu Asn Met
 405 410 415

<210> SEQ ID NO 5
 <211> LENGTH: 3573
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:

-continued

<223> OTHER INFORMATION: *Umbelopsis isabellina*

<400> SEQUENCE: 5

atgagcacca ccacctgat tgtggccatt aagggtaccg ataatgttga acgtgatggc	60
gcaattgttc gtctggatgc cggcgccagt atggaaaccc tgcgtccgcg cattgcagaa	120
aaactggcca ttagcagtgg cattgaagat ctgattcttg aagatgcaaa tggtgacaat	180
ctgaccacca ttgatcaggt tcgtaaacag cagaccgttt ttgttaatct ggaagatcag	240
attaagctgc cggcagttcc ggccataacc ctgccgtatt ttggtaatct gtatcagctg	300
ctgccggata tgctggccgg ttggcgcaaa ctgtttgatg aatatgggcc ggttggtaaa	360
gtgaatctgc tgggcaatga aattatttgt accaatgac cggccgtggc cgaactgtgg	420
gttaaagaaa gcgaatattt taccaaaaag atctacggtg gcctgcagga agttaaaagt	480
tttggcggcc agggctctgt taccaccgat agcgatgata tggattggaa actggcacat	540
aaactgctga tgcgggcatt ttcaccgcgc gccattaagg tttatcagca tgaaatgagc	600
gtgattggtc tgcagaccat taaggttttt gaacagtata gcccgatga agaagtggaa	660
attctgcatt ggaccaccaa tctgaccttt gaaaccattg gcaaagtgg ttttggtat	720
gattttcatc tgctggatga tcgttatggc gaaaatcatc cgttttattga agcaatgggt	780
tattgcatga aacagagctt tgcccggtgc acccagagca aactgattaa gtatctgccg	840
attgaagcca atcgctcgta tgatcgagct ctgaatctga tgcatagcac tgttgatgaa	900
gttattaccc agcgtaaaag ccataccgat gcaagcgaag ataataagga tctgctggat	960
tttatgctga ccgcacgtga tgaaaataat ctgggcctga gcgatgaaaa tattcgcgat	1020
caggttatta cctttctgat tgcggcccat gaaaccacca gcaataccct ggcatggacc	1080
ctgtatgaac tgagtcgcca tccggaaatt gaacagaaaa ttctgcagga agtgggtgaat	1140
ctgggcatta ccaccgattc actgccgacc agcgaacaga gcagtagcat gaaatatacc	1200
tatcaggtgc tgaagaagc cctgcgcgat tatagtcgcc tgcgtgcact ggcaaaaat	1260
tgcaaaaaag atattgttgt gccggggcgc tatcagatta aggcaggcga tcgtgtggcc	1320
gttcagctga atagcctgca ttataatgaa aaagtttacc cgaatccgac ccagtatgat	1380
ccgagtcgct ggaccccgga agaagaacag aaacgcagtc gctttgcctg gctgccgttt	1440
agcaccggcc cgcgtagctg catttgtatg gcaactggccc tgcaggaagc aaaaaccatt	1500
ctggcaatga ttctgctgaa atttcgcttt gtttatgatg gtccgccgat tggttatgat	1560
ccgaaaagcc cgaccattcg tccgctgaat ctgatgatga aaattctgcc gcgtgataaa	1620
ctgccgagcc cgaccgcga taataagctg accccggttg gcagtcgaa tatttgtaaa	1680
gccagatgg caatgccgac cgcagccacc aatgttggtg ccgcgaaact gccgcgggtt	1740
ttctttctgt atggtacca gaccggtacc gcccaggatt atgcaagtca gctggcaagc	1800
caggccaaaa gttttggttt taaagatatt accctgtgtg aaatggataa atgggaagtg	1860
ctgcagagcg gcaaatatac cgttgccaaa gaagaaaaag ataccctgac cctggttgtt	1920
atttgtaccg caacctataa tggcagcccg ccggatagtg cagaaaattt tgataaattc	1980
attaccgaca ccagcaaaga taatgatctg ccgctgaaag gcctgctgta taccgtgttt	2040
ggcctgggca atcgtaattg gcgcacctat cagcagtttc cgattaagtg cgatgcccg	2100
ctggatgaac tgggtgtgta acgcttttat gatctgggca gcggcaatgc cgataaagat	2160
atggatggty aatttcatga ttggtgcgcc catttttggg cccataccct gacctattat	2220
ggcattgcaa cccatgaagg tagtgcaagt ctggtgccgg gccagaaaa caccgaagat	2280

-continued

```

ccgaccagcg gtctggaaat tcgttttatt ccgccgagcg aaagcgaagc ctgggataaa 2340
gccgataaaa atgttaatgg tgactataat gcggaaatcc aggtgaatcg cgaactgcag 2400
aatattgaac gtagtaaacg tagcaccgcc catattgaaa ttgatattag taaactgcag 2460
agcccgata cggataaaca tcgttatctg accggtgacc atctggaagt gtatccggaa 2520
aatgccgata atgtggtgga aagcattgcc ctgggctttg gtctgattct ggatagtgtg 2580
tttgaaatta ccagcgtgga taaaagtacc gtgagtagtc gtagcctggc agccagtatt 2640
accggcccggt gtaccgttcg taatgcctcg aaatattatg ccgatata tagtgccccg 2700
agccgttacc tgctggcctt ttctgcagca cgctgcagg ataccatcc ggatgttgcc 2760
gcaagtttta gcgatgtgat tattccgggt gaaaatggcc agaaagccta taatgaatc 2820
attcagaaat accgtaacct gctggatctg cagcgcggct ttccgctgaa agaactggat 2880
ctgaaagaat ttctgtgcgc agttagcgtt atgcagccgc gccgttatag cattgccagc 2940
agcccgctga aagcagaaga taccgcatat ctggcagttg gtgtggtgga tgatgtgttt 3000
gcaaatactc attattatgg cctggccagt ggctatctgg ccgcagcca gccgccgaca 3060
ccgattcgtg cacgtattaa gagtagcaaa agtacctttg gcctgccgga aaatccggaa 3120
accccgatta ttatgattgg tgcgggtacc ggtattagtc cgtttatggg ctttatgcag 3180
gaacgcgaaa tgctgaaaag caaagcagaa gcccatctgt ttttcggctg ccgccatccg 3240
gatgaagatt ttatctatcg tagcgttttt gaaggctatg aaaaaagcgg cgttattacc 3300
aaactgtatc cggccttttag ccgtctgggt gaagataatc cgcgcaaata tgttcagcat 3360
cagctgatgg caaatgccgg caaaatttgg accctgctga ccaatggcgc aaatgtgtat 3420
gtttgcggca gtggcccgat gagcccgcat gtgcgtcgta gctttgaact gatggcaaaa 3480
agttttggag gcgcaaaaac cgatgatgaa gcatgcgaaa aactgctgga attcatgagc 3540
gcaggtcgtt ataataaaga tgtgtggggt taa 3573

```

```

<210> SEQ ID NO 6
<211> LENGTH: 1190
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Umbelopsis isabellina

```

```

<400> SEQUENCE: 6

```

```

Met Ser Thr Thr Thr Leu Ile Val Ala Ile Lys Gly Thr Asp Asn Val
1           5           10          15
Glu Arg Asp Gly Ala Ile Val Arg Leu Asp Ala Gly Ala Ser Met Glu
20          25          30
Thr Leu Arg Pro Arg Ile Ala Glu Lys Leu Ala Ile Ser Ser Gly Ile
35          40          45
Glu Asp Leu Ile Leu Glu Asp Ala Asn Gly Asp Asn Leu Thr Thr Ile
50          55          60
Asp Gln Val Arg Lys Gln Gln Thr Val Phe Val Asn Leu Glu Asp Gln
65          70          75          80
Ile Lys Leu Pro Ala Val Pro Ala His Thr Leu Pro Tyr Phe Gly Asn
85          90          95
Leu Tyr Gln Leu Leu Pro Asp Met Leu Ala Gly Trp Arg Lys Leu Phe
100         105         110
Asp Glu Tyr Gly Pro Val Val Lys Val Asn Leu Leu Gly Asn Glu Ile
115         120         125

```


Ile 130	Gly	Thr	Asn	Asp	Pro	Ala 135	Val	Ala	Glu	Leu	Trp 140	Val	Lys	Glu	Ser
Glu 145	Tyr	Phe	Thr	Lys	Lys 150	Ile	Tyr	Gly	Gly	Leu 155	Gln	Glu	Val	Lys	Ser 160
Phe	Gly	Gly	Gln	Gly 165	Leu	Phe	Thr	Thr	Asp 170	Ser	Asp	Asp	Met	Asp 175	Trp
Lys	Leu	Ala	His 180	Lys	Leu	Leu	Met	Pro 185	Ala	Phe	Ser	Pro	Arg 190	Ala	Ile
Lys	Val	Tyr 195	Gln	His	Glu	Met	Ser 200	Val	Ile	Gly	Leu	Gln 205	Thr	Ile	Lys
Val 210	Phe	Glu	Gln	Tyr	Ser	Pro 215	Asp	Glu	Glu	Val	Glu 220	Ile	Leu	His	Trp
Thr 225	Thr	Asn	Leu	Thr	Phe 230	Glu	Thr	Ile	Gly	Lys 235	Val	Gly	Phe	Gly	Tyr 240
Asp	Phe	His	Leu	Leu 245	Asp	Asp	Arg	Tyr	Gly 250	Glu	Asn	His	Pro	Phe 255	Ile
Glu	Ala	Met	Gly 260	Tyr	Cys	Met	Lys	Gln	Ser	Phe	Ala	Arg	Gly 270	Thr	Gln
Ser	Lys 275	Leu	Ile	Lys	Tyr	Leu	Pro 280	Ile	Glu	Ala	Asn	Arg 285	Arg	Tyr	Asp
Arg 290	Ser	Leu	Asn	Leu	Met	His 295	Ser	Ile	Val	Asp	Glu 300	Val	Ile	Thr	Gln
Arg 305	Lys	Ser	His	Pro	His 310	Ala	Ser	Glu	Asp	Asn 315	Lys	Asp	Leu	Leu	Asp 320
Phe	Met	Leu	Thr	Ala 325	Arg	Asp	Glu	Asn	Asn 330	Leu	Gly	Leu	Ser	Asp 335	Glu
Asn	Ile	Arg	Asp 340	Gln	Val	Ile	Thr	Phe 345	Leu	Ile	Ala	Gly 350	His	Glu	Thr
Thr	Ser 355	Asn	Thr	Leu	Ala	Trp 360	Thr	Leu	Tyr	Glu	Leu 365	Ser	Arg	His	Pro
Glu 370	Ile	Glu	Gln	Lys	Ile	Leu 375	Gln	Glu	Val	Val	Asn 380	Leu	Gly	Ile	Thr
Thr 385	Asp	Ser	Leu	Pro	Thr 390	Ser	Glu	Gln	Ser	Ser 395	Ser	Met	Lys	Tyr	Thr 400
Tyr	Gln	Val	Leu	Lys 405	Glu	Thr	Leu	Arg	Met 410	Tyr	Ser	Pro	Leu	Arg 415	Ala
Leu	Ala	Lys	Tyr 420	Cys	Lys	Lys	Asp	Ile	Val	Val	Pro	Gly 430	Gly	Tyr	Gln
Ile	Lys 435	Ala	Gly	Asp	Arg	Val	Ala 440	Val	Gln	Leu	Asn	Ser 445	Leu	His	Tyr
Asn 450	Glu	Lys	Val	Tyr	Pro	Asn 455	Pro	Thr	Gln	Tyr	Asp 460	Pro	Ser	Arg	Trp
Thr 465	Pro	Glu	Glu	Glu	Gln 470	Lys	Arg	Ser	Arg	Phe 475	Ala	Trp	Leu	Pro	Phe 480
Ser	Thr	Gly	Pro	Arg 485	Ser	Cys	Ile	Gly	Met 490	Ala	Leu	Ala	Leu	Gln 495	Glu
Ala	Lys	Thr	Ile 500	Leu	Ala	Met	Ile	Leu	Leu 505	Lys	Phe	Arg	Phe	Val	Tyr
Asp	Gly 515	Pro	Pro	Ile	Gly	Tyr	Asp 520	Pro	Lys	Ser	Pro	Thr 525	Ile	Arg	Pro
Leu 530	Asn	Leu	Met	Met	Lys	Ile 535	Leu	Pro	Arg	Asp	Lys 540	Leu	Pro	Ser	Pro
Thr	Ala	Asp	Asn	Lys	Leu	Thr	Pro	Val	Gly	Ser	Pro	Asn	Ile	Gly	Lys

-continued

545	550	555	560
Ala Gln Met	Ala Met Pro Thr	Ala Ala Thr Asn Val Gly Thr	Ala Glu
	565	570	575
Leu Pro Pro	Val Phe Phe Leu Tyr Gly Thr	Gln Thr Gly Thr	Ala Gln
	580	585	590
Asp Tyr Ala	Ser Gln Leu Ala Ser Gln Ala Lys Ser	Phe Gly Phe Lys	
	595	600	605
Asp Ile Thr	Leu Cys Glu Met Asp Lys Trp Glu Val	Leu Gln Ser Gly	
	610	615	620
Lys Tyr Thr	Gly Ala Lys Glu Glu Lys Asp Thr Arg	Ala Leu Val Val	
	625	630	635
Ile Cys Thr	Ala Thr Tyr Asn Gly Ser Pro Pro Asp Ser	Ala Glu Asn	
	645	650	655
Phe Asp Lys	Phe Ile Thr Asp Thr Ser Lys Asp Asn Asp	Leu Pro Leu	
	660	665	670
Lys Gly Leu	Leu Tyr Thr Val Phe Gly Leu Gly Asn Arg	Asn Trp Arg	
	675	680	685
Thr Tyr Gln	Gln Phe Pro Ile Lys Cys Asp Ala Arg	Leu Asp Glu Leu	
	690	695	700
Gly Gly Glu	Arg Phe Tyr Asp Leu Gly Ser Gly Asn Ala Asp	Lys Asp	
	705	710	715
Met Asp Gly	Glu Phe His Asp Trp Cys Ala His Phe Trp Thr	His Thr	
	725	730	735
Leu Thr Tyr	Tyr Gly Ile Ala Thr His Glu Gly Ser Ala Ser	Leu Val	
	740	745	750
Pro Gly Gln	Lys Thr Thr Glu Asp Pro Thr Ser Gly Leu Glu Ile Arg		
	755	760	765
Phe Ile Pro	Pro Ser Glu Ser Glu Ala Trp Asp Lys Ala Asp	Lys Asn	
	770	775	780
Val Asn Gly	Asp Tyr Asn Ala Glu Ile Gln Val Asn Arg Glu Leu Gln		
	785	790	795
Asn Ile Glu	Arg Ser Lys Arg Ser Thr Arg His Ile Glu Ile Asp Ile		
	805	810	815
Ser Lys Leu	Gln Ser Pro Asp Thr Asp Lys His Arg Tyr Leu Thr Gly		
	820	825	830
Asp His Leu	Glu Val Tyr Pro Glu Asn Ala Asp Asn Val Val Glu Ser		
	835	840	845
Ile Ala Leu	Gly Phe Gly Leu Ile Leu Asp Ser Val Phe Glu Ile Thr		
	850	855	860
Ser Val Asp	Lys Ser Thr Val Ser Ser Arg Ser Leu Ala Ala Ser Ile		
	865	870	875
Thr Gly Pro	Cys Thr Val Arg Asn Ala Leu Lys Tyr Tyr Ala Asp Ile		
	885	890	895
Tyr Ser Ala	Pro Ser Arg Tyr Leu Leu Ala Phe Phe Ala Ala Arg Leu		
	900	905	910
Gln Asp Thr	His Pro Asp Val Ala Ala Ser Phe Ser Asp Val Ile Ile		
	915	920	925
Pro Gly Glu	Asn Gly Gln Lys Ala Tyr Asn Glu Phe Ile Gln Lys Tyr		
	930	935	940
Arg Asn Leu	Leu Asp Leu Gln Arg Gly Phe Pro Leu Lys Glu Leu Asp		
	945	950	955
Leu Lys Glu	Phe Leu Cys Ala Val Ser Val Met Gln Pro Arg Arg Tyr		
	965	970	975

-continued

Ser Ile Ala Ser Ser Pro Leu Lys Ala Glu Asp Thr Ala Tyr Leu Ala
 980 985 990

Val Gly Val Val Asp Asp Val Phe Ala Asn Arg His Tyr Tyr Gly Leu
 995 1000 1005

Ala Ser Gly Tyr Leu Ala Arg Ser Gln Pro Pro Thr Pro Ile Arg
 1010 1015 1020

Ala Arg Ile Lys Ser Ser Lys Ser Thr Phe Gly Leu Pro Glu Asn
 1025 1030 1035

Pro Glu Thr Pro Ile Ile Met Ile Gly Ala Gly Thr Gly Ile Ser
 1040 1045 1050

Pro Phe Met Gly Phe Met Gln Glu Arg Glu Met Leu Lys Ser Lys
 1055 1060 1065

Ala Glu Ala His Leu Phe Phe Gly Cys Arg His Pro Asp Glu Asp
 1070 1075 1080

Phe Ile Tyr Arg Ser Val Phe Glu Gly Tyr Glu Lys Ser Gly Val
 1085 1090 1095

Ile Thr Lys Leu Tyr Pro Ala Phe Ser Arg Leu Gly Glu Asp Asn
 1100 1105 1110

Pro Arg Lys Tyr Val Gln His Gln Leu Met Ala Asn Ala Gly Lys
 1115 1120 1125

Ile Trp Thr Leu Leu Thr Asn Gly Ala Asn Val Tyr Val Cys Gly
 1130 1135 1140

Ser Gly Pro Met Ser Arg Asp Val Arg Arg Ser Phe Glu Leu Met
 1145 1150 1155

Ala Lys Ser Phe Gly Gly Ala Lys Thr Asp Asp Glu Ala Cys Glu
 1160 1165 1170

Lys Leu Leu Glu Phe Met Ser Ala Gly Arg Tyr Asn Glu Asp Val
 1175 1180 1185

Trp Gly
 1190

<210> SEQ ID NO 7
 <211> LENGTH: 30
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 7

ggaattccat atggaaagta aagtagttgt

30

<210> SEQ ID NO 8
 <211> LENGTH: 32
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Primer

<400> SEQUENCE: 8

ccgctcgagt tacatgtttt cgatgatcgc gt

32

What is claimed is:

1. A method of making lactones, the method comprising:

- (a) incubating a cellular system expressing a heterologous cytochrome P450 (CYP450) protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, wherein the straight chain fatty acid and the subterminal hydroxy fatty acid

have 5 to 40 carbon atoms, and wherein the heterologous cytochrome P450 protein comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of a cytochrome P450 protein from *Umbelopsis isabellina*, and either:

- (b1) incubating the subterminal hydroxy fatty acid with a yeast cell culture to produce lactones, wherein the

61

- lactones comprise at least one gamma-lactone, at least one delta-lactone, or combinations thereof, or
- (b2) incubating the subterminal hydroxy fatty acids in an acidic environment to produce lactones, wherein the lactones comprise at least one gamma-lactone, at least one delta-lactone, or combinations thereof.
2. The method of claim 1, wherein the heterologous cytochrome P450 protein comprises the amino acid sequence of SEQ ID NO: 6.
3. The method of claim 1, wherein the cellular system expresses an exogenous isocitrate dehydrogenase (Icd) protein besides expressing an heterogenous cytochrome 450 protein.
4. The method of claim 3, wherein the Icd protein comprises an amino acid sequence that is at least 95% identical to the amino acid sequence of an Icd protein from *E. coli*.
5. The method of claim 3, wherein the Icd protein comprises the amino acid sequence of SEQ ID NO: 4.
6. The method of claim 1, wherein the straight chain fatty acid and the hydroxy fatty acid have 7 carbons to 20 carbons.
7. The method of claim 1, wherein the straight chain fatty acid is lauric acid or tridecylic acid.
8. The method of claim 1, wherein the subterminal hydroxy fatty acid is a hydroxylated lauric acid, or a hydroxylated tridecylic acid.
9. The method of claim 1, wherein the cellular system comprises bacterial cells, yeast cells, or a combination thereof.
10. The method of claim 1, wherein the lactones comprise five carbons to eight carbons.
11. The method of claim 1, wherein the lactones comprise one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone.
12. The method of claim 1, wherein the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone.

62

13. The method of claim 1, wherein the straight chain fatty acid is lauric acid, the subterminal hydroxyl fatty acid comprises one or more of ω -1 hydroxy lauric acid, ω -2 hydroxy lauric acid, and ω -3 hydroxy lauric acid, and the lactones comprise one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone.
14. The method of claim 1, wherein the straight chain fatty acid is tridecylic acid, the subterminal hydroxy fatty acid comprises one or more of ω -1 hydroxy tridecylic acid, ω -2 hydroxy tridecylic acid, and ω -3 hydroxy tridecylic acid, and the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone.
15. The method of claim 1, wherein the subterminal hydroxy fatty acid is isolated from the cellular system prior to step (b1) or (b2).
16. The method of claim 1, further comprising isolating the lactones from the yeast cell culture.
17. A method of making lactones, the method comprising:
- (a) incubating a cellular system expressing a heterologous cytochrome P450 protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, wherein the heterologous cytochrome P450 protein comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of a cytochrome P450 protein from *Umbe-
lopsi isabellina*; and
- wherein the straight chain fatty acid and the subterminal hydroxy fatty acid have 5 to 40 carbon atoms;
- (b) incubating the subterminal hydroxy fatty acid from step (a) with a cell culture to produce subterminal hydroxy fatty acid with the length shorter than the subterminal hydroxy fatty acid from step (a); and
 - (c) incubating the subterminal hydroxy fatty acids obtained at the end of step (b) in an acidic environment to produce lactones, wherein the lactones comprise at least one gamma-lactone, at least one delta-lactone, or combinations thereof.

* * * * *